

The Science of Philanthropy Cleanup

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Abstract

Research has documented a large number of conditions that affect charitable giving practices. To what extent is the science of philanthropy self-correcting, and which findings on charitable giving hold up against the scrutiny of scientific criticism? We answer these questions with an empirical assessment of statistical irregularities, retractions and replications of published findings. We find that in research on giving retractions and replications are less common than in research in other fields, and that statistical errors are more common. About 60% of replications yield similar conclusions as preceding studies. Previous findings on material benefits and social rewards for giving are less likely to replicate, while findings on material costs and solicitation are more likely to replicate. In the interest of a robust knowledge base for fundraising, we call for more replications and we suggest that journals adopt registered reports - in principle acceptance of publication before results are known - as a solution to publication bias.

Correspondence, Contributions, and Open Science

Correspondence: r.bekkers@vu.nl. This paper relies on research plans presented previously (Bekkers, 2022a, 2024a) and includes results presented previously (Table 1: 2024b; replication prevalence: Bekkers, 2024c; p-value analysis: Littel, 2025). Contributor Role Taxonomy ([CRediT](#)) statement: Funding acquisition, Conceptualization, Methodology, Formal analysis, Visualization, Writing – original draft, Project administration and Supervision: RB; Writing - Review and Editing: RB, WA, AG; Software: RB and ML; Validation: ML and SKM; Investigation and Data curation: RB, ML, SKM, AG, HF. The authors did not use generative artificial intelligence in the creation of this paper. The authors acknowledge helpful suggestions from Michaela Neumayr on a previous version, help from Marlou Ramaekers with data curation, and from Ji Ma with software. Data and software used to produce the results reported in this paper as well as supplementary materials are available at <https://osf.io/5thj9>

Introduction

In 2012, Boyan Slat, a first year aerospace engineering student in the Netherlands, came up with a clever idea to remove pollution from the world's oceans: a ship could gather plastic waste floating around, and bring it back ashore so that it can be recycled. The world's oceans are a truly global collective good: they are crucial for planetary health, but no one's particular ownership or responsibility to protect. All creatures, great and small, benefit from less pollution and higher ocean water quality.

The current state of science is similar to the plastic waste problem in many ways. We all benefit from clear waters in science. But more than a decade after the Open Science Collaboration (2015) showed that half of replications of seminal studies show very different results than their predecessors, it is clear that science is in a knowledge production crisis. The production of scientific papers has grown at a strong pace, but the reliability of research has not increased: a recent massive attempt to replicate research findings in the social and behavioral sciences again found that half of previous research findings was not replicated (Tyner et al., 2026).

Similar to the ocean clean up initiative, the current paper illustrates the potential of a simple solution that can identify waste in the ocean of knowledge on why people give. The solution is to screen the large number of published findings, to identify studies that attempted to replicate previous findings, and compare their results: replications with similar results as preceding studies reflect robust regularities and reliable knowledge (see also Hartshorne & Schachner, 2012). Especially independent replications - attempts to replicate findings by other researchers - are informative. Similar to the concern that the ocean clean up does not eliminate the causes of the environmental crisis, this initiative is not a solution to the knowledge production crisis. But it does give readers – scientists and practitioners alike – a simple tool to evaluate the reliability of published research, and that's important for our collective scientific health.

An important source of knowledge in science generally are evidence syntheses, such as meta-analyses and narrative systematic reviews (Siddaway, Wood & Hedges, 2019). One systematic review of the literature on charitable giving (Bekkers & Wiepking, 2011) distinguished eight mechanisms driving donations. Each mechanism is a category of interventions, i.e., a set of levers that fundraisers can pull to attract donations. The regularities emerging from studies that have tested these interventions are that donors give more (often) when 1) they are aware of a clear need for donations; 2) when they are explicitly being asked to give; 3) when the material costs of giving are lower and the material benefits are higher; 4) when they care about the well-being of beneficiaries; 5) when they receive social benefits for giving and avoid social costs for not giving; 6) when they receive psychological benefits for giving and avoid psychological costs for not giving; 7) when the cause they support matches their values; and 8) when they perceive donations as more effective and efficient.

Previous research

With the current paper we aim to raise the bar for what counts as reliable evidence in research on charitable giving. To guide our attempts to locate reliable findings, we rely on findings from the newly emerged fields

of meta science, research on research, and bibliometrics. Here we give a brief overview of the relevant literature.

First, we broadly define *replications* as attempts to repeat findings from previous studies. This definition is not limited to hypothesis-testing research, and includes exploratory and descriptive research. Replications come in 4 varieties: 1) verification: researchers analyze the same data as a preceding study, with the same methods and models; 2) robustness analyses: re-analysis of the same data with different methods and models; 3) reproduction: researchers analyze new data with the same methods and models; and 4) generalization: analysis of new data with different methods and models (Freese & Peterson, 2017).

In a recent attempt to reanalyze findings from 600 papers published in 62 journals in the social and behavioral sciences (Miske et al., 2026), researchers could locate data and code for only 144 papers. Reproductions with new data often produce different results (Tyner et al., 2026). The literature review on mechanisms driving charitable giving that we revisit here (Bekkers & Wiepking, 2011) described some features of the populations and contexts, but did not include a quality appraisal of included studies.

Meta science findings (Open Science Collaboration, 2015; Holzmeister et al., 2024; Youyou, Yang & Uzzi, 2023) suggest three categories of reasons why replications find different results than preceding studies. A first category of reasons we label *unique findings*. Studies reporting findings in unique contexts, for specific regions or subgroups of populations, or in unusual circumstances are unlikely to be encountered again when researchers attempt to do the study again in other times, regions, populations and circumstances. A second category of reasons is the *quality of preceding studies*. Studies containing statistical errors are less likely to be replicated. Studies reporting weaker effect sizes, with smaller numbers of observations and lower statistical power are less likely to be successfully reproduced in new data. A third category of reasons is the process of *publication, peer review, and citation*. Studies based on weak evidence but reporting significantly positive findings - i.e., supporting the hypotheses - are more likely to be published and cited than studies reporting null-findings or negative results. Such biases distort our view of truly reliable research, and create the impression that research is more replicable than it actually is.

1. *Unique findings*

There may be good reasons in the realities we study that explain why repeating a study can produce entirely different results. Preceding findings may reflect uniquely local patterns that do not reflect general regularities in human behavior. Testing the same hypothesis with other data can produce radically different results because the association truly differs between samples drawn from other populations. For charitable giving, interventions that generally tend to work in experiments among participants in lab studies or surveys does not guarantee positive results in each and every campaign fundraisers organize. The effects may be very different in practice. In addition, peculiarities of donor groups, public perceptions of fundraising organizations, and the causes that they raise donations for will create heterogeneity in treatment effects. Below we investigate to what extent replications produce results that differ from preceding findings as a result of differences in study populations and contexts.

2. Research quality

Honest mistakes. Sometimes researchers make errors, for example in copying results from statistical software to the word processors they use to write manuscripts. This is one source of the inconsistencies between reported p -values and test statistics that we quantify below. Sometimes researchers also forget to include details in their manuscripts that are crucial for the results. The academic community tolerates honest mistakes, especially if authors step forward to identify and correct them. Error detection and correction is a sign of integrity.

Researcher degrees of freedom. Replication failures and weaker associations in replications may also be related to researcher degrees of freedom in preceding studies (Simmons, Nelson & Simonsohn, 2011; Gelman & Loken, 2014). Data collection and analysis involve a large number of decisions: the choice of measurement instruments, treatment of outliers, and other data preprocessing choices, the selection, application and interpretation of analytical models and statistical tests. Fiddling with the analysis to obtain significant results is known as *p-hacking* (Simmons, Nelson, & Simonsohn, 2011). In the presence of incentives to report strong, significant and positive results, researchers will attempt to obtain them, and withhold negative results and null-findings. Gross inconsistencies in p -values are more common in statistical tests reported as significant than in tests reported as not significant, suggesting a systematic bias favoring positive results (Nuijten et al., 2016). Below we explore how often p -values in research on charitable giving are inconsistent with test statistics, and whether they contain a similar bias toward positive results.

3. The publication process

Published research is more often reporting positive results than unpublished research. Many relevant research findings never see the light of day, and those that stay in the shadows tend to tell us mostly which interventions are *not* working. Studies published in journals are more likely to report positive findings than studies that remain unpublished, as a result of author, editor, and reviewer behaviors (Ferguson & Heene, 2012; Franco et al., 2014; O’Boyle et al, 2017; Chopra et al., 2024). In economics, authors at all levels of seniority as well as editors believe that significant results submitted to journals are more likely to get published (Chopra et al., 2024). A survey in multiple fields suggests that authors are more likely to submit significant results for publication, and both reviewers and editors prefer them (Muradchianian et al., 2023).

The excess of positive results supporting the hypotheses in published research – approaching 95% in some disciplines (Fanelli, 2010, 2012; Schäfer & Schwarz, 2019) indicates that negative or null results are withheld from publication in file drawers (Rosenthal, 1979). Publication bias implies an asymmetry in the reliability of findings: positive findings are less reliable than negative findings. If there is no study reporting a theoretically plausible effect, it is very unlikely that the effect actually exists. The grains of salt added in the process of publication imply that reported findings cannot be trusted if no one attempted to replicate them.

A recurrent finding in replications is that they tend to produce statistically significant associations much less often than their predecessors. Replication studies in the social and behavioral sciences find different results about half of the time (Open Science Collaboration, 2015; Camerer et al., 2018; Tyner et al., 2026). Replications also tend to find substantially weaker effect sizes than the studies they attempt to replicate, so

that meta-analytic effects based solely on published research are likely inflated (Ioannidis, Stanley, & Doucouliagos, 2017; Kvarven, Strømland & Johannesson, 2020).

Evidence syntheses that summarize results of previous research on charitable giving in systematic literature reviews and meta-analyses typically report findings in articles published in scientific journals. As a result, the available systematic literature reviews are likely to provide a distorted picture: because negative findings are obscured, the conclusions of review articles are biased toward positive results. We do not know how strong publication bias is in research on charitable giving. Systematic literature reviews that include unpublished studies typically do not analyze differences between published and unpublished studies. In the analyses below we explore how often results in studies on giving are supportive of the study hypotheses, and whether the tendency to report positive results differs between replications and preceding studies, as well as between studies published in journals and those that remained unpublished.

A commonly used proxy for research quality is publication status: a study published in a journal or book that employs peer review is commonly believed to be ‘better’ than an unpublished study. Unfortunately, the peer review system of quality control is not fortified enough to keep out the bad science (Smith, 2010). Journals continue to rely on unpaid volunteers for reviews, and typically do not specify how reviewers should evaluate manuscripts, nor provide training in how to detect bad science. Studies that compared manuscripts submitted and accepted for publication (Stephen, 2022; Hopewell et al., 2014) found that they do not change much during the review period, indicating that the added value of review is small. Even the most prestigious journals have trouble keeping the bad science out of their pages (Brembs, 2018). As a result, the publication of a paper is no guarantee that the research is reliable, even when it is in *Science* or *Nature* (Camerer et al., 2018). The clearest cases of this filtering failure are *retractions*, where the flaws in a study become apparent only after publication. Below we examine how often, and for what reasons, research on charitable giving is retracted; we then identify, review and analyze replication studies in this field.

Data and Methods

This section briefly presents the data and methods we used to identify and code (1) retractions; (2) consistency of test statistics, and (3) replications. See Appendix A for details.

Retractions. To identify retractions of research on charitable giving, we (RB) searched Google Scholar, using the six keywords (“charitable behavior”, “charitable giving”, “prosocial behavior”, “philanthropy”, “altruism”, and “donations”) from Bekkers & Wiepking (2011) plus “retract*”.

Test statistics. To estimate the prevalence of inconsistencies in test statistics associated with p-values in research on giving, we applied the Statcheck module (Nuijten & Epskamp, 2024) integrated in metacheck (Lakens & DeBruine, 2026) to a large corpus of papers included in OpenAlex. First, one of the authors (RB) defined the corpus to include all studies in OpenAlex that cited at least one out of 24 seed publications (see Appendix B) reflecting the eight mechanisms in giving behavior described in Bekkers & Wiepking (2011). Next, one of the authors (ML) compiled a corpus of 7,412 studies included in OpenAlex citing these seed publications, and downloaded the largest possible number of publications from the corpus through Zotero, obtaining pdf files for 3,702 publications. Next, we (ML) ran Statcheck on these pdfs. Because Statcheck can

only process papers that adhere to journal article reporting standards of the American Psychological Association (APA), it could not process all papers.

Replications. To identify replications, we developed a method to identify replications from large volumes of research in two stages. In the first stage, we (RB) created a coding scheme and manually screened 1,213 Google Scholar search results for “charitable behavior” for terms related to replication and identified 34 replications. 25 replications analyzed charitable giving as a dependent variable, representing 0.5% of all 4,858 search results for charitable behavior. In the second stage, we (HF, AG, RB) modified this method to screen a more targeted selection of 340 search results for “charitable giving” mentioning replication, identifying another 28 replications analyzing charitable giving as a dependent variable. Finally, we added 13 replications to the analysis that had not shown up in our search results but that we knew existed.

Results

The generic increase in the number of publications researchers produce also holds in the field of research on philanthropy and charitable giving. The number of studies that researchers produce each year is very large and has grown strongly in the past decades (see Appendix A). The comprehensive overview of all studies on conditions that affect charitable giving published in 2007 (Bekkers & Wiepking, 2007) included about 500 studies, assembled from seven keyword searches in Google Scholar that yielded less than 5,000 search results. Due to increasing productivity in science, repeating this effort for studies that appeared after 2007 would be humanly impossible. Meanwhile, thousands of studies on charitable giving appear each year. In 2021, no less than 89,000 studies on donations appeared according to Google Scholar. For the same year, 35,300 publications on altruism appeared, 27,000 on philanthropy, 17,100 on prosocial behavior, 4,270 on charitable giving, and 467 on charitable behavior. It has become virtually impossible to keep track of the current state of knowledge. Since the year 2000, when the number of studies on charitable behavior was only 40 and on prosocial behavior 1,580, the number of publications has grown more than tenfold.

How often is research on giving retracted?

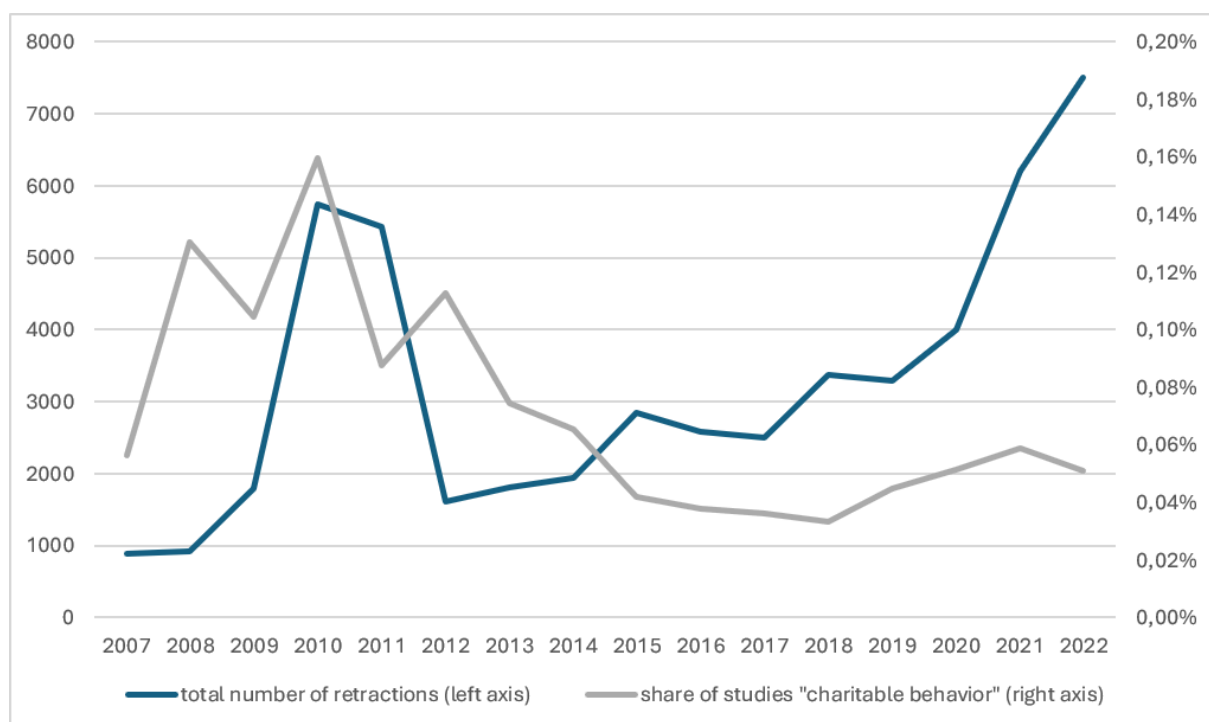
In Google Scholar we searched for retracted publications on charitable giving, using the six keywords from the mechanisms review plus “retract*” to identify studies that mentioned retracted, retracting, or retraction. We found 36 studies on charitable giving published between 2007 and 2022 that publishers retracted. We found no retracted studies on giving that were published before the year 2007.

In science overall, the total number of retracted studies in the database of Retraction Watch (2024) for the same period was 52,464. The retraction rate of studies on charitable giving was 0.069%. The number of retractions per year has increased strongly over the years. For studies on charitable giving, in contrast, we see a downward trend after the peak in 2010 (see Figure 1). For the period until 2014, studies on charitable giving are 0.010% of all retracted studies; for the period after 2014, they are 0.044%.

Two authors (RB and SKM) coded the retracted studies. 24 retracted studies (62%) were published in journals, the remaining 15 were conference abstracts. 26 retracted studies (67%) were empirical papers, the

remaining 13 were not reporting analyses of data. The majority of retracted empirical studies on giving ($n = 22$, 85%) did not analyze individual charitable giving as a dependent variable. Instead, these studies analyzed giving intentions, perceptions of the behavior of others, or corporate social responsibility. We located 31 of the 39 retracted studies in the Retraction Watch database, which contained the country of the first author, and the stated reason for retraction. The majority of retracted studies on charitable giving were from China ($n = 17$, 55%), similar to the proportion for the entire database of retractions (52%). Almost half (14 of 31) retracted studies on giving stated ‘peer review concerns’ or ‘breach of policy by author’ as a reason for retraction. In the entire database of retractions, these reasons are much less common (17% of all cases).

Figure 1: Total number of retractions per year of publication, and the retraction rate for studies on charitable giving



All retracted empirical studies that analyzed charitable giving as a dependent variable were published in psychology. Three of them – Chatterjee et al., (2013), Sanna et al. (2011) and Wang et al. (2012) – were retracted for fraudulent data. The retraction of Decety et al. (2015) deserves mentioning because the authors requested it after a reanalysis of the study data by other authors (Shariff et al., 2016) produced different results. The preceding findings were due to a coding error. In this case, independent scrutiny of data shared by the authors enabled the correction of the study findings.

Most retracted studies were rarely cited, if ever: in Google Scholar, 15 retracted publications had citation counts of 0. Three quarters were cited less than 20 times. In all, studies on charitable behavior are rarely retracted; retracted studies are rarely cited, and the retraction rate is declining. These findings may be

interpreted two fundamentally different ways. One interpretation is that research on charitable giving is trustworthy and reliable. Apparently publishers find little cause for retraction of research on giving. Another option is that research on giving has not received much scrutiny, and is therefore less likely to be retracted. Retractions tell us something about the peer review process. The flaws in the retracted studies were serious enough to warrant withdrawal, but reviewers did not catch them before publication. As stated above, almost half of the retractions stated "peer review concerns" or "breach of policy by author" as the reason, compared to only 17% in the entire database. Peer review apparently does not reliably keep bad science out, and in some cases, the problem lies in the review process or in author misconduct.

Availability of data greatly facilitates error detection. Yet transparency about data in the field of research on charitable giving is low. In 2023, less than half of empirical research papers in five journals (Voluntary Sector Review, Nonprofit Management and Leadership, Voluntas, NVSQ, and the Journal of Philanthropy and Marketing) provided any references to the data analyzed (Bekkers, 2023). Articles in NML and NVSQ were most likely to contain links to datafiles or information about data availability, but such information was given in only one out of five articles.

How often does research on charitable giving misreport p-values?

Of 667 publications that Statcheck could process, 306 (46%) produced some form of error. In 86 papers (13%) decision errors were found, where reported significance levels were different from the computed significance levels implied by the test statistics. Papers typically report multiple p-values. Of all 6,999 p-values checked, 776 (11%) were errors, and 128 were decision errors (1.8%). Both proportions are rather high compared to previous estimates for communication science (8.8% and 1.3%, respectively) and social psychology (8.7% and 1.5%, respectively; Vermeulen et al., 2015). Also they are higher than for psychology as a whole (9.7% and 1.4%, respectively; Nuijten et al., 2016).

89 of the 128 decision errors (70%) were favoring the study hypotheses, such that the authors incorrectly claimed a statistically significant relationship, when the test statistics actually indicated a non-significant relationship. This proportion is also higher than in communication science (65%; Vermeulen et al., 2015).

We do not have a good explanation why the error rates in studies on giving are higher than in (social) psychology and communication science, and why a higher proportion of the errors are favoring study hypotheses. These two findings may reflect both honest mistakes as well as researcher degrees of freedom. Honest mistakes may have occurred in transcribing results from software to manuscripts. Researchers studying charitable giving may have received less training in statistics. It is also possible that studies on giving contain more errors because peer reviewers and editors are less able to detect such errors. The pronounced skew toward errors that favor the hypotheses suggests researcher degrees of freedom.

What characterizes studies attempting to replicate previous findings?

In research on charitable giving, replications are rare: they represent less than 1% of the total number of publications, which is less than in most other disciplines of the social and behavioral sciences (Bekkers,

2024c). Among the 66 studies that replicated studies on charitable giving as a dependent variable, 39 studies (59%) found results that were similar to the preceding study, 27 did not: 17 studies reported null findings, and 10 studies reported findings in contrast to the preceding study. These results are similar to results of a recent attempt to replicate a large number of findings from the social and behavioral sciences (Tyner et al., 2026).

Unique findings. The majority of studies that sought to replicate a previous study on charitable behavior attempted to generalize the findings by using different data and conducting a different analysis than the preceding study ($n = 41$, 64%). Among these studies, the majority of replication results ($n = 25$, 61%) supported the preceding findings. Among the 25 studies that attempted to replicate a previous finding with either the same data or the same method, a similar proportion produced findings in line with previous results ($n = 14$, 56%). The difference is not significant ($\chi^2 = .159$, $df = 1$, $p = .690$).

Publication process. 48 studies that attempted to replicate preceding findings on charitable giving were published in journal articles or book chapters (73%), 18 were not: 7 were dissertations, master theses or bachelor theses; 11 were working papers, preprints, or conference contributions. Among the 39 studies replicating previous findings that found results in line with the replicated study, a clear majority ($n = 29$, 74%) was published. Among the 27 studies that found null or negative results, a similar proportion of studies ($n = 19$, 70%) was published. The difference is not significant ($\chi^2 = .128$, $df = 1$, $p = .721$).

What can we learn from replications of research on giving?

We present the reliable findings from replications by the primary mechanism examined in Table 1. Among the 66 studies on charitable behavior and charitable giving that attempted to replicate previous findings (see Appendix C) 60 examined at least one of the eight mechanisms. Attention to the mechanisms was unevenly distributed: material costs and benefits (19 replications), psychological costs and benefits (15), and efficacy (7) were most likely to be the primary mechanism examined.

In the replications we identified, there is no reliable evidence supporting the regularities as stated in Bekkers & Wiepking (2011) on awareness of need, altruism, reputation, and values. The lack of reliable evidence for these mechanisms is not a result of a lack of research. There are plenty of studies, but there are no replications with similar results as the claims presented in Bekkers & Wiepking (2011) from studies that measured donations.

Awareness of need. The highly cited “identifiable victim effect” (Jenni & Loewenstein, 1997), suggests that people find it easier to process information about the needs of a single individual than about the needs of a large number of individuals. Yet replications in field experiments (Lesner & Rasmussen, 2014; Thomas-Walters & Raihani, 2017) do not find evidence for the original claim that opportunities to donate to a single, identifiable victim produce more donations than opportunities to donate to multiple, statistical victims. A recent registered report (Vu et al., 2024) also did not find support for the identified victim effect. In one field experiment, contemplation of needs of victims in prayer did not increase donations (Greenway, Schnitker, & Shepherd, 2018).

Table 1. Reliable findings from replications of studies on charitable giving, by primary mechanism

Mechanism	n	Reliable findings
Awareness of need	3	-
Solicitation	4	More giving after direct verbal requests (Andreoni et al., 2017)
Material costs and benefits	15	More giving when the price of giving is lower (Davis et al., 2005), though tax-price effects are smaller in France (Fack & Landais, 2010), Germany (Huck & Rasul, 2011), and the UK (Almunia et al., 2020) <i>Less</i> giving when donors receive material benefits (Naeem & Zaman, 2009, 2016; Newman & Shen, 2012)
Altruism	5	Most donors do <i>not</i> give for purely altruistic reasons (Gangullia et al., 2020)
Reputation	6	Cues of being watched do <i>not</i> increase giving (Saunders et al., 2016; Ekström, 2012) Public recognition did <i>not</i> affect donations (Jung et al, 2023; Denis, Pecheux, & Warlop, 2020) or <i>reduced</i> donations (Lambarraa & Riener, 2015)
Psychological benefits	19	More giving when price reductions are presented as matches instead of rebates (Bekkers, 2015; Davis et al., 2005; Sasaki, Kurokawa & Ohtake, 2022) More giving when people think about one unit first, and how many to give next (Marcinkiewicz, 2016; Diederich et al., 2021) Depicting a single, identified victim does <i>not</i> raise more money than multiple, statistical victims (Hart, Lane & Chinn, 2018) Sense of control explains why giving time is more attractive than giving money (Costillo & Malkoc, 2022)
Values	1	-
Efficacy	7	More giving when the overhead costs of fundraising organizations are lower (Gneezy et al., 2014) More giving when donations are more likely to help reach a fundraising target (Verhaert & Van den Poel, 2012)

Altruism. Results of replication studies show that most people do not donate for purely altruistic reasons. In laboratory experiments with very clear instructions to students show that some of them indeed allocate funds provided to them in ways that maximize the well-being of others (Gandullia et al., 2020) as one would predict from pure altruism (Ottoni-Wilhelm, Vesterlund, & Xie, 2017). However, in ordinary situations, most donors do not respond to changes in the need for contributions when nonprofit organizations lose or gain government grants (De Wit & Bekkers, 2017, 2020). When customers can benefit a charity at no cost to themselves, they tend to do so, but they do not pay more to yield a larger benefit for the charity (Jung et al., 2017). A large field experiment in Alaska appealing to benefits for recipients (List et al., 2021) found that they raised less donations than appeals to the benefits for donors.

Reputation. The often cited result from an experiment among a lab study among a small sample of students that cues of being watched would increase donations (Haley & Fessler, 2005) is not reliable. A field experiment in Japan found it only when previous donations were larger (Oda & Ichihashi, 2016). It did not emerge from multiple follow-up experiments among larger numbers of participants, with null-findings in both online experiments (Bekkers, 2011; Saunders et al., 2016) as well as in field experiments (Ekström, 2012; Jung et al., 2023).

The claim that public recognition increases donations was neither replicated in a large set of field experiments in US shops (Jung et al, 2023), nor in an online experiment among Facebook users in France (Denis, Pecheux, & Warlop, 2020). Announcing that in-kind donations of books would be publicized did not increase donations in a field experiment in the UK (Cotterill, John & Richardson, 2013). Observability of donations did not increase gifts in a field experiment in cinemas in Germany (Dannenberg, Johansson-Stenman & Wetzel, 2022). Observability *reduced* giving among muslims in Morocco (Lamberaa & Riener, 2015), a finding that the authors attribute to giving norms discouraging public display of charity. In a registered report replicating a study on donor preferences, participants in an online experiment preferred charities that would keep donations anonymous (Chan & Feldman, 2025).

Values. There is strong evidence that they are correlated with donation behavior (Bekkers & Ottoni-Wilhelm, 2016; Nilsson, Erlandsson & Västfjäll, 2020). Yet most field experiments - an exception is Grodeck & Schoenegger (2023) - have not found positive effects of moral appeals on donation behavior (Frohly et al., 2026). This pattern of results suggests that people with more prosocial values give more, but that reminding people of prosocial values does not make them give more. The highly cited finding that higher social class individuals display lower levels of giving behavior (Piff et al., 2010), potentially as a result of higher greed, narcissism and lower prosocial values, has met with failures to replicate (Korndörfer, Egloff & Schmukle, 2015; Stamos et al., 2020).

We did find some evidence supporting the regularities on solicitation, material costs and benefits, psychological benefits, and efficacy. At the same time, we also found evidence qualifying or in two cases even consistently rejecting the regularities reported in preceding studies.

Solicitation. Direct solicitations yielded more donations in two large field experiments in the US. At the same time, the same experiments showed that a quarter of persons did not open the door (Dellavigna, List & Malmendier, 2012). About one third of consumers in a supermarket in Boston avoided exposure to fundraisers (Andreoni, Rao & Trachtman, 2017) when they knew they would ask them to donate. The latter finding was not replicated in Alaska, where entering the supermarket through another door exposed shoppers to more extreme cold (Trachtman et al., 2015). Opera house donors in Germany gave 42% less when they knew a second solicitation would be coming (Adena & Huck, 2019). Students in Switzerland gave 33% less when solicited twice vs once (Grieder, Schmitz & Schubert, 2021). In a field experiment in Denmark, sending reminders to regular donors increased donations, but also unsubscriptions (Damgaard & Gravert, 2018). An analysis of panel survey data in the Netherlands (Bekkers, 2026) revealed that conversion rates of direct solicitations declined strongly in the past 20 years.

Material costs and benefits. For material costs, we found clear support of the regularity that price reductions encourage donations (Bekkers, 2015; Davis et al., 2005; Zhang, 2023). Tax price studies (Fack & Landais, 2010; Almunia et al., 2020) and experiments (Huck & Rasul, 2011) tend to find smaller estimates than in previous research (Karlan & List, 2007; Peloza & Steel, 2005). For material benefits, a study in Pakistan that sought to repeat preceding findings that material benefits would encourage donations actually found the opposite result: unconditional gifts reduced giving (Naeem & Zaman, 2009, published in 2016). An experiment on thank you gifts found that they *reduced* donations among existing donors in the US (Newman & Shen, 2012; also see Fang & Bian, 2022).

Psychological costs and benefits. In a ‘stress-test’ of preceding findings (Eckel & Grossman, 2003), the framing as price reductions in the form of matches by third parties rather than as rebates produced more giving in a replication study (Davis et al., 2005). A field test with earned money among survey participants in the Netherlands found that matches attract more donations than rebates (Bekkers, 2015). Similar results emerged from a study in Japan (Sasaki et al., 2022). Finally, a field experiment in Germany among donors to an opera house (Huck & Rasul, 2011) also found that matches increase donations.

The finding that people give more when they think about one unit first, and how many to give next (Hsee et al., 2013) found support in two replications (Marcinkiewicz, 2016; Diederich et al., 2022; also see Diederich et al., 2023; Epperson et al., 2025). Two studies measuring donations in hypothetical situations (Karlsson et al., 2020; Maier et al., 2023) also found support for the unit asking effect.

A series of replications (Kessler & Meier, 2014) suggest that effects of cognitive load manipulations (Loewenstein & Small, 2007) on giving are limited to situations in which people are not exposed to other information.

Effects of mortality salience on giving have been much weaker in independent replications (Bruine de Bruin & Ulqinako, 2021; Roberts & Maxfield, 2019) than previously reported (Jonas, Schimel, Greenberg & Pyszczynski, 2002). One study in Norway measuring donation preferences found no effect (Sætrevik & Sjøstad, 2022).

Field experiments with social information treatments have resulted in much smaller (Van Teunenbroek & Bekkers, 2019) or no effects (Murphy et al., 2015) than previously reported (Croson & Shang, 2008; Shang & Croson, 2009).

Efficacy. In line with preceding results, replications have found that donors are responsive to information about the overhead costs of fundraising organizations, and give more when overhead costs are lower (Gneezy, Keenan & Gneezy, 2014; Portillo & Stinn, 2018; Charles, Sloan & Schubert, 2020). This finding also emerges from a study measuring willingness to donate (Hung, Berrett, & Ma, 2025). In addition, one experiment attempting to replicate findings on seed money - an intervention successfully applied by Boyan Slat to fund the Ocean Cleanup - found that participants give more often when their donations are more likely to help reach a fundraising target (Verhaert & Van den Poel, 2012).

Discussion

Limitations: what are we missing in this analysis?

To identify the largest possible number of replications, we used Google Scholar because this is the most comprehensive database of research (Martín-Martín et al., 2021). Compared to other databases such as Scopus and OpenAlex, Google Scholar includes research reports from a larger variety of sources such as dissertations, working papers, and preprints. The results indicate that these sources are particularly valuable because they tend to report less positive results. Relying on less comprehensive databases, we would have failed to find them, thereby sustaining the biases in the publication process that create the illusion that most studies support the hypotheses.

Despite the comprehensiveness of our search strategy, the findings reported above are limited to a relatively small proportion of the entire body of published research on conditions that affect charitable giving. There will be studies that attempt to repeat previous findings on conditions that are affecting giving behavior that we did not find with the keywords “charitable behavior” and “charitable giving”.

Furthermore, because our search strategy only produced studies that explicitly mentioned terms related to replication of research, the results may not fully represent the population of studies that can be considered to repeat previous findings. Authors repeating previous research may not always present their studies as such, especially if they work in a field where novelty has a high premium. In addition, when their results are different from the preceding study, they can anticipate pushback from the preceding study authors if they were to be invited to review the repeated study findings.

Both limitations imply that our overview is incomplete. It is quite likely that we are missing insights on giving just because they used slightly different terms. Furthermore, we did not estimate effect sizes in replications and preceding studies. Our findings on publication bias have sobering implications for systematic reviews and meta-analyses that rely solely on published research. They should be read with caution.

In our assessment of replications, we did not quantify effect sizes and compare them with the replicated studies. Our impression is that effect sizes in replication studies are smaller than in the studies they replicated, as in mass replication efforts (Open Science Collaboration, 2015; Camerer et al., 2018; Tyner et al., 2026). Particularly in studies testing the mechanisms of solicitation, material costs, reputation, and psychological costs and benefits replications found weaker effects than preceding studies.

In the analysis of p-values we were limited by the software we used. We could only analyze studies for which we could download the full text and to manuscripts that adhered to APA reporting guidelines. Also, we could not analyze p-values from tables. Ideally we would also have analyzed p-values reported in unpublished research, and in journals that do not require compliance with APA guidelines. The error rate among all studies is likely to be higher than among those that we could analyze.

Future research: What's next?

It will not come as a surprise that the first recommendation for future research is that the current study should be repeated. The current results do not represent the entire field of research on philanthropy. A complete science of philanthropy cleanup should revisit all research on philanthropy, including studies on the organization and the effects of philanthropy. The field of research on giving is growing so fast that it is no longer feasible to collect, summarize, review, and analyze all studies by hand. Future work can use computational social science tools (Ma et al., 2023) to automate the identification and classification of replications. The database that we created for the current project by manually coded features of replications and preceding studies can serve as training data in the development of rule-based algorithms.

Our second recommendation is that researchers should replicate findings on charitable giving more frequently and consistently, and should actively look for replications of promising findings before acting upon them in practice. Without the replications covered in the current paper, readers would have continued to believe that watching eyes, public recognition of donations, identified victims, and thank you gifts increase donations.

A third recommendation is that future evidence syntheses should compare effect sizes in replications and preceding studies. In our sample, published studies did not present more positive results than unpublished studies. However, replications tended to find weaker main effects than the studies they replicated. To learn which interventions have reliable effects, it is important to have the complete picture. Registered reports are a solution to this problem. After a stringent peer review of the thoroughness of the design and methods, authors of registered reports can be sure of publication of their findings, regardless of whether they are positive, negative, or null. In psychology, only 44% of registered reports provide support for the study hypothesis, while 96% of traditional studies support the hypothesis (Scheel, Schijen & Lakens, 2021).

A fourth recommendation is that peer review of research on charitable giving should enhance the scrutiny of statistical results. The field of nonprofit and philanthropy research has recently begun to discuss open science practices that may help to improve the reliability of knowledge (Bekkers et al., 2025). *Nonprofit & Voluntary Sector Quarterly* introduced a data and code review policy for manuscripts reporting results from experiments, requiring computational verification that the data and code produce the results reported in the manuscript (Bekkers, 2020). We recommend that journals publishing research based on other sources of data should also scrutinize numerical data and code before publication. In the disciplines that are more often sharing data and code, replication rates are higher (Miske et al., 2026). Our analyses of inconsistencies in p-values indicated that almost half of published studies we could process include errors. Even though authors, reviewers, editors, and journal production staff can easily identify them, these errors were not caught in peer review. This kind of error is just one example of a large number of questionable research practices that pollute the body of knowledge on giving. Using metacheck (DeBruine & Lakens, 2026) at the preprint stage or in peer review can prevent these errors.

Conclusion

We find that both retractions and replications are rare in research on conditions that affect charitable giving. Furthermore, we find that reporting errors in statistical tests in research on charitable giving are more common than in other fields, and that they are more often favoring the hypotheses. Even if the errors are honest mistakes, their prevalence indicates that quality control mechanisms in research on charitable giving are working less effectively than in other fields.

We answered our question which interventions have reliable effects on charitable giving behavior by a rigorous removal of all studies for which we have no evidence from replications by other scholars confirming the preceding results. We found that about 60% of replications yield similar conclusions as preceding studies, in line with recent replication attempts in the social and behavioral sciences (Tyner et al., 2026). Reductions of material costs, exposing people to direct solicitations, unit asking, matching, and lower overhead costs have reliably positive results in independent replications. We find that published findings on reputation and material benefits in charitable giving do not hold up against the scrutiny of scientific criticism through replications. Finally, we find no evidence for publication bias in replications. It is good to know that the science of philanthropy is self-correcting: researchers published replication failures, for instance with respect to reputation and material benefits. Increasing the scrutiny of research on charitable giving and publishing replications regardless of results in registered reports will further improve the robustness and reliability of research on charitable giving.

References

References to replications analyzed are in Appendix C

Aria, M., Le T., Cuccurullo, C., Belfiore, A. & Choe, J. (2024). openalexR: An R-Tool for Collecting Bibliometric Data from OpenAlex. *The R Journal*, 15(4), 167-180. <https://doi.org/10.32614/RJ-2023-089>.

Bekkers, R. (2020). A Data Transparency Policy for Results Based on Experiments. July 8, 2020. <https://renebekkers.wordpress.com/2020/07/08/a-data-transparency-policy-for-results-based-on-experiments/>

Bekkers, R. (2022a). Stained glass: A Critical Assessment of Transparency in Nonprofit Research. Research plan presented at the 51st ARNOVA Conference, November 18, 2022. <https://osf.io/4re7g>

Bekkers, R. (2022b). Ten Meta Science Insights to Deal With the Credibility Crisis in the Social Sciences. SocArxiv Preprint, <https://osf.io/preprints/socarxiv/rm4p8/>

Bekkers, R. (2023). Transparency in Nonprofit and Philanthropy Research. 11th ERNOP Conference, Zagreb, 29 June 2023. <https://osf.io/kqd2j>

Bekkers, R. (2024a). Eight Mechanisms, Seventeen Years Later. November 7, 2024. <https://renebekkers.wordpress.com/2024/11/07/eight-mechanisms-seventeen-years-later/>

Bekkers, R. (2024b). What We Really Know About Why People Give. ERNOP Webinar, December 11, 2024. <https://osf.io/jegvd>

Bekkers, R. (2024c). How often do we replicate previous research? <https://renebekkers.wordpress.com/2024/03/08/how-often-do-we-replicate-previous-research/> March 8, 2024.

Bekkers, R. (2026). Prosocial values for the philanthropic commons. In: Understanding our Philanthropic Commons, edited by Brenda Bushouse, Brent Never, and Robert Christensen. Cambridge University Press. <https://osf.io/mwkca/files/uyv3b>

Bekkers, R., Chapman, C., Prosser, A., Wiepking, P. & Meyer, M. (2025). Let us be clear: Why and how to improve data and methods transparency in nonprofit research. *Voluntas*, 36(4): 473 - 482. <https://doi.org/10.1007/s11266-025-00736-0>

Bekkers, R. & Ottoni-Wilhelm, M. (2016). Principle of Care and Giving to Help People in Need. *European Journal of Personality*, 30(3): 240-257. <http://dx.doi.org/10.1002/per.2057>

Bekkers, R. & Wiepking, P. (2007). Generosity and Philanthropy: A Literature Review. Report commissioned by the John Templeton Foundation. https://renebekkers.files.wordpress.com/2011/08/bekkers_wiepking_generosity_07.pdf

- Bekkers, R. & Wiepking, P. (2011). A Literature Review of Empirical Studies of Philanthropy: Eight Mechanisms that Drive Charitable Giving. *Nonprofit and Voluntary Sector Quarterly*, 40(5): 924-973. <https://doi.org/10.1177/0899764010380927>
- Boyce, V., Mathur, M., & Frank, M. C. (2023). Eleven years of student replication projects provide evidence on the correlates of replicability in psychology. *Royal Society Open Science*, 10(11), 231240. <https://doi.org/10.1098/rsos.231240>
- Brembs, B. (2018). Prestigious Science Journals Struggle to Reach Even Average Reliability. *Frontiers of Human Neuroscience*, 12(37): 1-7. <https://doi.org/10.3389/fnhum.2018.00037>
- Camerer, C. F., Dreber, A., Holzmeister, F., Ho, T. H., Huber, J., Johannesson, M., ... & Wu, H. (2018). Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015. *Nature Human Behaviour*, 2(9), 637-644. <https://doi.org/10.1038/s41562-018-0399-z>
- Chan, M., & Feldman, G. (2025). Factors impacting effective altruism: Revisiting heuristics and biases in charity in a replication and extension registered report of Baron and Szymanska (2011). *Royal Society Open Science*, 12(5), 250290. <https://doi.org/10.1098/rsos.250290>
- Chatterjee, P., Rose, R. L., & Sinha, J. (2013). RETRACTED ARTICLE: Why money meanings matter in decisions to donate time and money. *Marketing Letters*, 24, 109-118. <https://doi.org/10.1007/s11002-012-9215-0>
- Chopra, F., Haaland, I., Roth, C., & Stegmann, A. (2024). The null result penalty. *The Economic Journal*, 134(657), 193-219. <https://doi.org/10.1093/ej/uead060>
- Cotterill, S., John, P., & Richardson, L. (2013). The Impact of a Pledge Request and the Promise of Publicity: A Randomized Controlled Trial of Charitable Donations. *Social Science Quarterly*, 94(1), 200–216. <https://doi.org/10.1111/j.1540-6237.2012.00896.x>
- Croson, R., & Shang, J. Y. (2008). The impact of downward social information on contribution decisions. *Experimental Economics*, 11(3), 221-233. <https://doi.org/10.1007/s10683-007-9191-z>
- Decety, J., Cowell, J. M., Lee, K., Mahasneh, R., Malcolm-Smith, S., Selcuk, B., & Zhou, X. (2015). RETRACTED: The Negative Association between Religiousness and Children's Altruism across the World. *Current Biology*, 25(22), 2951-2955. [https://www.cell.com/current-biology/fulltext/S0960-9822\(15\)01167-7](https://www.cell.com/current-biology/fulltext/S0960-9822(15)01167-7)
- DellaVigna, S., List, J. A., & Malmendier, U. (2012). Testing for altruism and social pressure in charitable giving. *Quarterly Journal of Economics*, 127(1), 1-56. <https://doi.org/10.1093/qje/qjr050>
- Denis, E., Pecheux, C., & Warlop, L. (2020). When public recognition inhibits prosocial behavior: The case of charitable giving. *Nonprofit and Voluntary Sector Quarterly*, 49(5), 951-968. <https://doi.org/10.1177/0899764020911203>

- De Wit, A., & Bekkers, R. (2017). Government support and charitable donations: A meta-analysis of the crowding-out hypothesis. *Journal of Public Administration Research and Theory*, 27(2), 301-319. <https://doi.org/10.1093/jopart/muw044>
- De Wit, A., & Bekkers, R. (2020). Can charitable donations compensate for a reduction in government funding? The role of information. *Public Administration Review*, 80(2), 294-304. <https://doi.org/10.1111/puar.13154>
- Diederich, J., Epperson, R., and Goeschl, T. (2023). How to design the ask? Funding units vs. giving money. AWI Discussion Paper No. 698, Heidelberg University. https://archiv.ub.uni-heidelberg.de/volltextserver/33827/1/Diederich_Epperson_Goeschl_2023_dp731.pdf
- Eckel, C. C., & Grossman, P. J. (2003). Rebate versus matching: does how we subsidize charitable contributions matter? *Journal of Public Economics*, 87(3-4), 681-701. [https://doi.org/10.1016/S0047-2727\(01\)00094-9](https://doi.org/10.1016/S0047-2727(01)00094-9)
- Epperson, R., Diederich, J., & Goeschl, T. (2025). How to Design the Ask? Funding Units vs. Giving Money. *Management Science*, 71(4), 2830-2846. <https://doi.org/10.1287/mnsc.2021.00157>
- Fanelli, D. (2010). “Positive” results increase down the hierarchy of the sciences. *PloS One*, 5(4), e10068. <https://doi.org/10.1371/journal.pone.0010068>
- Fanelli, D. (2012). Negative results are disappearing from most disciplines and countries. *Scientometrics*, 90(3), 891-904. <https://doi.org/10.1007/s11192-011-0494-7>
- Ferguson, C. J., & Heene, M. (2012). A vast graveyard of undead theories: Publication bias and psychological science’s aversion to the null. *Perspectives on Psychological Science*, 7(6), 555-561. <https://doi.org/10.1177/1745691612459059>
- Franco, A., Malhotra, N., & Simonovits, G. (2014). Publication bias in the social sciences: Unlocking the file drawer. *Science*, 345(6203), 1502-1505. <https://doi.org/10.1126/science.1255484>
- Freese, J., & Peterson, D. (2017). Replication in social science. *Annual Review of Sociology*, 43, 147-165. <https://doi.org/10.1146/annurev-soc-060116-053450>
- Frohly, C., Galbiati, R., Henry, E., & Jacquemet, N. (2026). Forced to face the truth: A meta-analysis on the effectiveness of moral reminders. <https://shs.hal.science/halshs-05456784/>
- Gelman, A., & Loken, E. (2014). The statistical crisis in science data-dependent analysis—a “garden of forking paths”—explains why many statistically significant comparisons don’t hold up. *American Scientist*, 102(6), 460-465. <https://www.jstor.org/stable/43707868>
- Jonas, E., Schimel, J., Greenberg, J., & Pyszczynski, T. (2002). The Scrooge effect: Evidence that mortality salience increases prosocial attitudes and behavior. *Personality and Social Psychology Bulletin*, 28(10), 1342-1353. <https://doi.org/10.1177/014616702236834>

- Korndörfer, M., Egloff, B., & Schmukle, S. C. (2015). A large scale test of the effect of social class on prosocial behavior. *PLoS One*, 10(7), e0133193. <https://doi.org/10.1371/journal.pone.0133193>
- Gneezy, U., Keenan, E. A., & Gneezy, A. (2014). Avoiding overhead aversion in charity. *Science*, 346(6209), 632-635. <https://doi.org/10.1126/science.1253932>
- Grodeck, B., & Schoenegger, P. (2023). Demanding the morally demanding: Experimental evidence on the effects of moral arguments and moral demandingness on charitable giving. *Journal of Behavioral and Experimental Economics*, 103, 101988. <https://doi.org/10.1016/j.socec.2023.101988>
- Harbaugh, W. T. (1998). The prestige motive for making charitable transfers. *American Economic Review*, 88(2), 277-282. <https://www.jstor.org/stable/116933>
- Hart, P.S., Lane, D., & Chinn, S. (2018). The elusive power of the individual victim: Failure to find a difference in the effectiveness of charitable appeals focused on one compared to many victims. *PLoS ONE*, 13(7): e0199535. <https://doi.org/10.1371/journal.pone.0199535>
- Hartshorne, J. K., & Schachner, A. (2012). Tracking replicability as a method of post-publication open evaluation. *Frontiers in Computational Neuroscience*. 6, 8, <https://doi.org/10.3389/fncom.2012.00008>
- Holzmeister, F., Johannesson, M., Böhm, R., Dreber, A., Huber, J., & Kirchler, M. (2024). Heterogeneity in effect size estimates. *Proceedings of the National Academy of Sciences*, 121(32), e2403490121. <https://doi.org/10.1073/pnas.2403490121>
- Hopewell, S., Collins, G. S., Boutron, I., Yu, L. M., Cook, J., Shanyinde, M., ... & Altman, D. G. (2014). Impact of peer review on reports of randomised trials published in open peer review journals: retrospective before and after study. *BMJ*, 349. <https://doi.org/10.1136/bmj.g4145>
- Hsee, C. K., Zhang, J., Lu, Z. Y., & Xu, F. (2013). Unit asking: A method to boost donations and beyond. *Psychological Science*, 24(9), 1801-1808. <https://doi.org/10.1177/0956797613482947>
- Ioannidis, J., Stanley, T. D., & Doucouliagos, H. (2017). The Power of Bias in Economics Research. *Economic Journal*, 127(605): F236-F265. <https://doi.org/10.1111/ecoj.12461>
- Jenni, K. E., & Loewenstein, G. F. (1997). Explaining the Identifiable Victim Effect. *Journal of Risk and Uncertainty*, 14, 235–257. <https://doi.org/10.1023/A:1007740225484>
- Kühberger, A., Fritz, A., & Scherndl, T. (2014). Publication bias in psychology: A diagnosis based on the correlation between effect size and sample size. *PLoS One*, 9(9), e105825. <https://doi.org/10.1371/journal.pone.0105825>
- Kvarven, A., Ström, E. & Johannesson, M. (2020). Comparing meta-analyses and preregistered multiple-laboratory replication projects. *Nature Human Behavior*, 4: 423-434. <https://doi.org/10.1038/s41562-019-0787-z>
- Lakens, D. & DeBruine, L. (2026). metacheck: Check Scientific Papers for Best Practices. <https://github.com/scienceverse/metacheck>

- Lambarraa, F., & Riener, G. (2015). On the norms of charitable giving in Islam: Two field experiments in Morocco. *Journal of Economic Behavior & Organization*, 118, 69-84.
<https://doi.org/10.1016/j.jebo.2015.05.006>
- Lesner, T. H., & Rasmussen, O. D. (2014). The identifiable victim effect in charitable giving: evidence from a natural field experiment. *Applied Economics*, 46(36), 4409-4430.
<https://doi.org/10.1080/00036846.2014.962226>
- List, J. A., Murphy, J. J., Price, M. K., & James, A. G. (2021). An experimental test of fundraising appeals targeting donor and recipient benefits. *Nature Human Behaviour*, 5(10), 1339-1348.
<https://doi.org/10.1038/s41562-021-01095-8>
- Littel, M.J. (2025). Misinterpretation of p-values in philanthropy research. <https://osf.io/9v5km>
- Loewenstein, G., & Small, D. A. (2007). The scarecrow and the tin man: The vicissitudes of human sympathy and caring. *Review of General Psychology*, 11(2), 112-126.
<https://doi.org/10.1037/1089-2680.11.2.112>
- Ma, J., Ebeid, I.A., De Wit, A., Xu, M., Yang, Y., Bekkers, R. & Wiepking, P. (2023). Computational Social Science for Nonprofit Studies: Developing a Toolbox and Knowledge Base for the Field. *Voluntas*, 34: 52-63.
<https://osf.io/g9d8u/>
- Martín-Martín, A., Thelwall, M., Orduna-Malea, E., & Delgado López-Cózar, E. (2021). Google Scholar, Microsoft Academic, Scopus, Dimensions, Web of Science, and OpenCitations' COCI: a multidisciplinary comparison of coverage via citations. *Scientometrics*, 126(1), 871-906.
<https://doi.org/10.1007/s11192-020-03690-4>
- Miske et al. (2026). Investigating the reproducibility of the social and behavioural sciences. *Nature*, 652: 126–134. <https://doi.org/10.1038/s41586-026-10203-5>
- Karlsson, H., Hellström, S., Moche, H., & Västfjäll, D. (2020). Unit Asking—a method for increasing donations: A replication and extension. *Judgment and Decision Making*, 15(6), 989-993.
<https://doi.org/10.1017/S1930297500008184>
- Muradchanian, J., Hoekstra, R., Kiers, H., & van Ravenzwaaij, D. (2023). The role of results in deciding to publish: A direct comparison across authors, reviewers, and editors based on an online survey. *PLoS One*, 18(10), e0292279. <https://doi.org/10.1371/journal.pone.0292279>
- Nilsson, A., Erlandsson, A., & Västfjäll, D. (2020). Moral foundations theory and the psychology of charitable giving. *European Journal of Personality*, 34(3), 431-447.
<https://journals.sagepub.com/doi/pdf/10.1002/per.2256>
- Nuijten, M. B. & Epskamp, S. (2024). statcheck: Extract statistics from articles and recompute p-values. R package version 1.5.0. Web implementation at <https://statcheck.io>

- Nuijten, M. B., Hartgerink, C. H., Van Assen, M. A., Epskamp, S., & Wicherts, J. M. (2016). The prevalence of statistical reporting errors in psychology (1985–2013). *Behavior Research Methods*, 48(4), 1205-1226. <https://doi.org/10.3758/s13428-015-0664-2>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251). <https://doi.org/10.1126/science.aac4716>
- Ottoni-Wilhelm, M., Vesterlund, L., & Xie, H. (2017). Why do people give? Testing pure and impure altruism. *American Economic Review*, 107(11), 3617-3633. <https://doi.org/10.1257/aer.20141222>
- Piff, P. K., Kraus, M. W., Côté, S., Cheng, B. H., & Keltner, D. (2010). Having less, giving more: the influence of social class on prosocial behavior. *Journal of Personality and Social Psychology*, 99(5), 771-784. <https://doi.org/10.1037/a0020092>
- Portillo, J. E., & Stinn, J. (2018). Overhead aversion: Do some types of overhead matter more than others? *Journal of Behavioral and Experimental Economics*, 72, 40-50. <https://doi.org/10.1016/j.socec.2017.11.003>
- Retraction Watch (2024). Retraction Watch Database. New York: the Center for Research Integrity. Downloaded on 4 November 2024. <https://gitlab.com/crossref/retraction-watch-data>
- Rosenthal, R. (1979). The file drawer problem and tolerance for null results. *Psychological Bulletin*, 86(3), 638-641. <https://doi.org/10.1037/0033-2909.86.3.638>
- Sætrevik, B., & Sjøstad, H. (2022). Mortality salience effects fail to replicate in traditional and novel measures. *Meta-psychology*, 6. <https://doi.org/10.15626/MP.2020.2628>
- Sanna, L. J., Chang, E. C., Miceli, P. M., & Lundberg, K. B. (2011). RETRACTED: Rising up to higher virtues: Experiencing elevated physical height uplifts prosocial actions. *Journal of Experimental Social Psychology*, 47, 472–476. <https://doi.org/10.1016/j.jesp.2010.12.013>
- Schäfer, T., & Schwarz, M. A. (2019). The meaningfulness of effect sizes in psychological research: Differences between sub-disciplines and the impact of potential biases. *Frontiers in Psychology*, 10, 813. <https://doi.org/10.3389/fpsyg.2019.00813>
- Scheel, A. M., Schijen, M. R., & Lakens, D. (2021). An excess of positive results: Comparing the standard Psychology literature with Registered Reports. *Advances in Methods and Practices in Psychological Science*, 4(2). <https://doi.org/10.1177/25152459211007467>
- Shang, J., & Croson, R. (2009). A field experiment in charitable contribution: The impact of social information on the voluntary provision of public goods. *The Economic Journal*, 119(540), 1422-1439. <https://doi.org/10.1111/j.1468-0297.2009.02267.x>
- Shariff, A. F., Willard, A. K., Muthukrishna, M., Kramer, S. R., & Henrich, J. (2016). What is the association between religious affiliation and children's altruism?. *Current Biology*, 26(15), R699-R700. <https://doi.org/10.1016/j.cub.2016.06.031>

Siddaway, A. P., Wood, A. M., & Hedges, L. V. (2019). How to do a systematic review: a best practice guide for conducting and reporting narrative reviews, meta-analyses, and meta-syntheses. *Annual Review of Psychology*, 70(1), 747-770. <https://doi.org/10.1146/annurev-psych-010418-102803>

Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11), 1359-1366. <https://doi.org/10.1177/0956797611417632>

Smith, R. (2010). Classical peer review: an empty gun. *Breast Cancer Research*, 12 (S4), S13. <https://doi.org/10.1186/bcr2742>

Stamos, A., Lange, F., Huang, S. C., & Dewitte, S. (2020). Having less, giving more? Two preregistered replications of the relationship between social class and prosocial behavior. *Journal of Research in Personality*, 84, 103902. <https://doi.org/10.1016/j.jrp.2019.103902>

Stephen, D. (2022). Peer reviewers equally critique theory, method, and writing, with limited effect on the final content of accepted manuscripts. *Scientometrics*, 127(6), 3413-3435. <https://doi.org/10.1007/s11192-022-04357-y>

Trachtman, H., Steinkruger, A., Wood, M., Wooster, A., Andreoni, J., Murphy, J. J., & Rao, J. M. (2015). Fair weather avoidance: Unpacking the costs and benefits of “avoiding the ask”. *Journal of the Economic Science Association*, 1(1), 8-14. <https://doi.org/10.1007/s40881-015-0006-2>

Tyner, A.H., Abatayo, A.L., Daley, M., et al. (2026). Investigating the replicability of the social and behavioural sciences. *Nature*, 652: 143–150. <https://doi.org/10.1038/s41586-025-10078-y>

Vermeulen, I., Beukeboom, C. J., Batenburg, A., Avramiea, A., Stoyanov, D., van de Velde, B., & Oegema, D. (2015). Blinded by the light: How a focus on statistical “significance” may cause p-value misreporting and an excess of p-values just below .05 in communication science. *Communication Methods and Measures*, 9(4), 253-279. <https://doi.org/10.1080/19312458.2015.1096333>

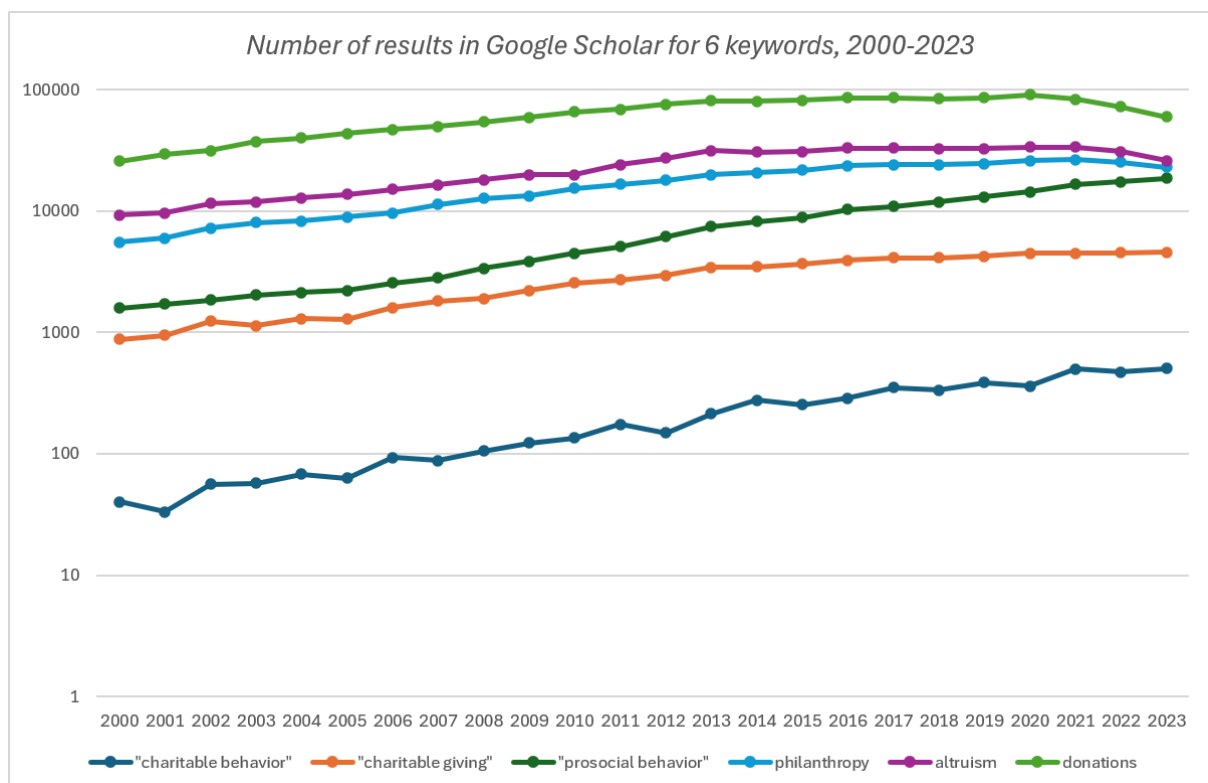
Wang, C. C., Cheng, Y. Y., Chiou, W. B., & Kung, C. C. (2012). Shame for money: Shame enhances the incentive value of economic resources—RETRACTED. *Judgment and Decision Making*, 7(1), 77-85. <https://doi.org/10.1017/S1930297500001856>

Youyou, W., Yang, Y., & Uzzi, B. (2023). A discipline-wide investigation of the replicability of Psychology papers over the past two decades. *Proceedings of the National Academy of Sciences*, 120(6), e2208863120. <https://doi.org/10.1073/pnas.2208863120>

Appendix A. Data and Methods details

To identify the largest possible number of replications and retractions of studies on charitable giving, we searched Google Scholar. We used the exact same search terms as in Bekkers & Wiepking (2011): “charitable behavior”, “charitable giving”, “prosocial behavior”, “philanthropy”, “altruism”, and “donations”. Figure A1 shows the growth of the number of search results for these 6 keywords in the period 2000 - 2023 in searches conducted in November 2024. For “charitable behavior”, the increase from 40 results in 2000 to 503 in 2023 represents a growth of 1258%. For “charitable giving”, the growth from 522 in 2000 to 4,570 in 2023 is 522%. Strong growth is also evident for “prosocial behavior” from 1,580 in 2000 to 18,600 in 2023 (1177%). For the other keywords growth rates are somewhat smaller: 413% for “philanthropy”, 280% for “altruism”, and 231% for “donations”. In all years, the largest set of results emerged for “donations”, exceeding 90,000 in 2020. The number of results for “donations” amounted to 1.5 million results over the period 2000-2023. The number of search results for the keywords “philanthropy”, “altruism” and “donations” declined somewhat after 2020, while “charitable behavior” and “charitable giving” continued to grow. We note that discrepancies will arise when these searches are conducted again because Google continuously updates the Scholar database, adding newly indexed records and reordering the ranks of search results.

Figure A1. Number of results in Google Scholar for 6 keywords, 2000-2023



The figure shows the number of search results in Google Scholar for six of the seven keywords used in Bekkers & Wiepking (2007; 2011). Note that the vertical axis is displayed on a logarithmic (base 10) scale.

Source: search conducted on 4 November 2024, reported in Bekkers (2024b), <https://osf.io/5thj9/files/jxkgs>.

Identifying retractions of studies on charitable giving

To identify studies on charitable giving that were retracted, we (RB) searched Google Scholar in November 2024 for the 6 keywords, adding “retracted” and “retraction”. These searches produced a total number of $n = 204,000$ records, including duplicates. For each search, RB manually screened the titles and abstracts displayed, until three pages of results in a row did not include retracted studies. Links to search strings are on tab “GS Results” and metadata of retracted studies are on tab “All” of <https://osf.io/5thj9/files/ywvth>.

Assessments of retracted studies (by SKM and RB) are in <https://osf.io/5thj9/files/k3gwn>. On 4 November 2024 we downloaded the Retraction Watch database and matched the retracted studies identified through the Google Scholar search. The Stata do file <https://osf.io/5thj9/files/jrucw> produces the results reported in the manuscript and additional results. The output log including all results is <https://osf.io/5thj9/files/xdrc3>.

Identifying inconsistent test statistics and decision errors in research on charitable giving

We fully report the data and methods used to identify inconsistencies in test statistics reported in research on charitable giving in Littel (2025), <https://osf.io/9v5km>. We (RB) created a stratified purposive sample of 24 articles, reflecting 2 to 4 key publications for each of the eight mechanisms in charitable giving Bekkers and Wiepking (2011). We provide the list of seed articles below. Next, we (ML) used the snowball function of the ROpenalex package (Aria et al., 2024) and identified 7412 papers that cited at least one of the seed articles. The pdfs of 3702 of these papers could be downloaded automatically by Zotero. We (ML) applied the R package metacheck (DeBruine & Lakens, 2026) to convert the pdfs into XML files, enabling targeted search for test statistics and assessment of their consistency with the statchek module (Nuijten & Epskamp, 2024). A limitation of statchek is that it can only process manuscripts adhering to reporting guidelines for statistical tests issued by the American Psychological Association. The APA guidelines are used in journals within the discipline of psychology and related fields. General interest journals such as *Science*, the *Proceedings of the National Academy of Sciences (PNAS)*, *Nature*, *Nature Human Behavior*, and journals of disciplines other than psychology, notably economics, public administration, biology, and sociology, do not require authors to report statistical tests according to the APA guidelines. Articles in these journals are more likely to report p-values in tables with symbols such as asterisks (*) for various significance levels. Statchek cannot process p-values reported in this manner. As a result, statchek could only analyze a minority of studies ($n = 659$, 18%) citing one of the seed articles.

In the pdfs in which statchek detected precise p-values and test statistics, 306 (46.4%) contained some kind of erroneous test statistic or p-value. An error was classified as a decision error when it made the difference between being above or below the common .05 threshold. For example, if the reported p-value is .045 while the computed p-value is .055, this constitutes a decision error since one of them is statistically significant at the conventional $p = .05$ threshold, while the other is not. A decision error was classified as favoring the study hypothesis when the reported result of the statistical supports the hypothesis while the computed p-value does not.

Appendix B. Seed articles for citation search in OpenAlex

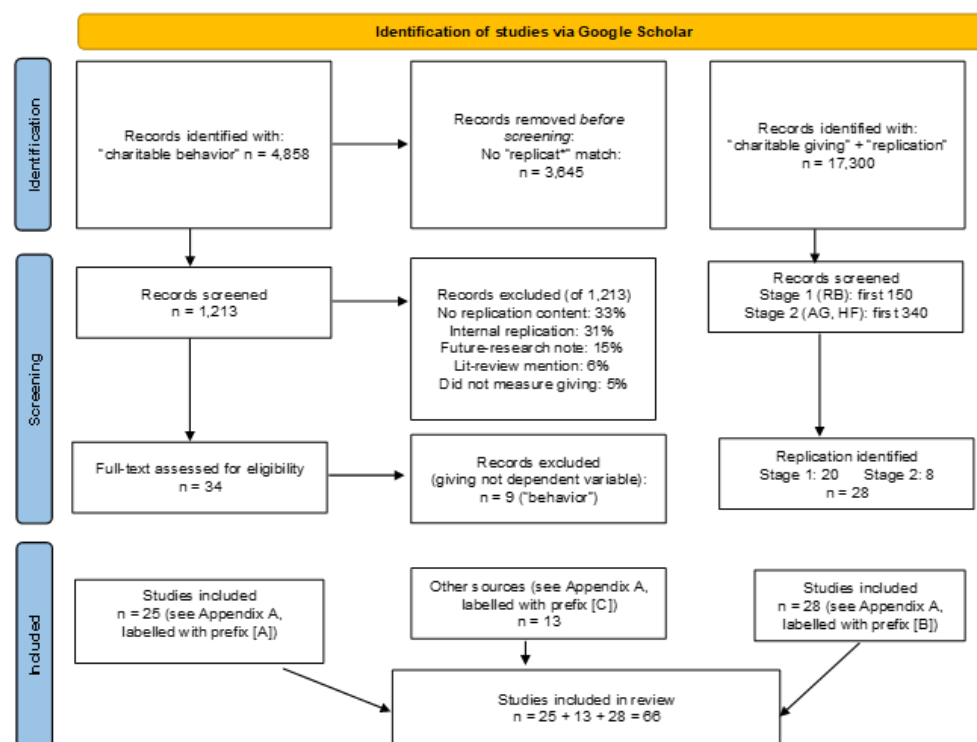
- Alpizar, F., Carlsson, F., & Johansson-Stenman, O. (2008). Does context matter more for hypothetical than for actual contributions? Evidence from a natural field experiment. *Experimental Economics*, 11(3), 299–314. <https://doi.org/10.1007/s10683-007-9194-9>
- Basil, D. Z., Ridgway, N. M., & Basil, M. D. (2006). Guilt appeals: The mediating effect of responsibility. *Psychology and Marketing*, 23(12), 1035–1054. <https://doi.org/10.1002/mar.20145>
- Bennett, R. (2003). Factors underlying the inclination to donate to particular types of charity. *International Journal of Nonprofit and Voluntary Sector Marketing*, 8(1), 12–29. <https://doi.org/10.1002/nvsm.198>
- Dolinski, D., Grzyb, T., Olejnik, J., Prusakowski, S., & Urban, K. (2005). Let's Dialogue About Penny: Effectiveness of Dialogue Involvement and Legitimizing Paltry Contribution Techniques. *Journal of Applied Social Psychology*, 35(6), 1150–1170. <https://doi.org/10.1111/j.1559-1816.2005.tb02164.x>
- Doob, A. N., & McLaughlin, D. S. (1989). Ask and You Shall be Given: Request Size and Donations to a Good Cause. *Journal of Applied Social Psychology*, 19(12), 1049–1056. <https://doi.org/10.1111/j.1559-1816.1989.tb01238.x>
- Fong, C. M. (2007). Evidence from an Experiment on Charity to Welfare Recipients: Reciprocity, Altruism and the Empathic Responsiveness Hypothesis. *The Economic Journal*, 117(522), 1008–1024. <https://doi.org/10.1111/j.1468-0297.2007.02076.x>
- Freedman, J. L., & Fraser, S. C. (1966). Compliance without pressure: The foot-in-the-door technique. *Journal of Personality and Social Psychology*, 4(2), 195–202. <https://doi.org/10.1037/h0023552>
- Haley, K. J., & Fessler, D. M. T. (2005). Nobody's watching? Evolution and Human Behavior, 26(3), 245–256. <https://doi.org/10.1016/j.evolhumbehav.2005.01.002>
- Harbaugh, W. T., Mayr, U., & Burghart, D. R. (2007). Neural Responses to Taxation and Voluntary Giving Reveal Motives for Charitable Donations. *Science*, 316(5831), 1622–1625. <https://doi.org/10.1126/science.1140738>
- Jackson, N. C., & Mathews, R. M. (1995). Using Public Feedback to Increase Contributions to a Multipurpose Senior Center. *Journal of Applied Behavior Analysis*, 28(4), 449–455. <https://doi.org/10.1901/jaba.1995.28-449>
- Karlan, D., & List, J. A. (2007). Does Price Matter in Charitable Giving? Evidence from a Large-Scale Natural Field Experiment. *American Economic Review*, 97(5), 1774–1793. <https://doi.org/10.1257/aer.97.5.1774>
- Kingma, B. R. (1989). An Accurate Measurement of the Crowd-out Effect, Income Effect, and Price Effect for Charitable Contributions. *Journal of Political Economy*, 97(5), 1197–1207. <https://doi.org/10.1086/261649>

- Lindskold, S., Forte, R. A., Haake, C. S., & Schmidt, E. K. (1977). The Effects of Directness of Face-to-Face Requests and Sex of Solicitor on Streetcorner Donations. *The Journal of Social Psychology*, 101(1), 45–51. <https://doi.org/10.1080/00224545.1977.9923982>
- List, J. A., & Lucking-Reiley, D. (2002). The Effects of Seed Money and Refunds on Charitable Giving: Experimental Evidence from a University Capital Campaign. *Journal of Political Economy*, 110(1), 215–233. <https://doi.org/10.1086/324392>
- List, J. A., Murphy, J. J., Price, M. K., & James, A. G. (2021). An experimental test of fundraising appeals targeting donor and recipient benefits. *Nature Human Behaviour*, 5(10), 1339–1348. <https://doi.org/10.1038/s41562-021-01095-8>
- Okten, C., & Weisbrod, B. A. (2000). Determinants of donations in private nonprofit markets. *Journal of Public Economics*, 75(2), 255–272. [https://doi.org/10.1016/S0047-2727\(99\)00066-3](https://doi.org/10.1016/S0047-2727(99)00066-3)
- Pancer, S. M., McMullen, L. M., Kabatoff, R. A., Johnson, K. G., & Pond, C. A. (1979). Conflict and avoidance in the helping situation. *Journal of Personality and Social Psychology*, 37(8), 1406–1411. <https://doi.org/10.1037/0022-3514.37.8.1406>
- Parsons, L. M. (2007). The Impact of Financial Information and Voluntary Disclosures on Contributions to Not-For-Profit Organizations. *Behavioral Research in Accounting*, 19(1), 179–196. <https://doi.org/10.2308/bria.2007.19.1.179>
- Payne, A. A. (1998). Does the government crowd-out private donations? New evidence from a sample of non-profit firms. *Journal of Public Economics*, 69(3), 323–345. [https://doi.org/10.1016/S0047-2727\(98\)00005-X](https://doi.org/10.1016/S0047-2727(98)00005-X)
- Pelozo, J., & Steel, P. (2005). The Price Elasticities of Charitable Contributions: A Meta-Analysis. *Journal of Public Policy & Marketing*, 24(2), 260–272. <https://doi.org/10.1509/jppm.2005.24.2.260>
- Satow, K. L. (1975). Social approval and helping. *Journal of Experimental Social Psychology*, 11(6), 501–509. [https://doi.org/10.1016/0022-1031\(75\)90001-3](https://doi.org/10.1016/0022-1031(75)90001-3)
- Small, D. A., & Simonsohn, U. (2008). Friends of Victims: Personal Experience and Prosocial Behavior. *Journal of Consumer Research*, 35(3), 532–542. <https://doi.org/10.1086/527268>
- Wagner, C., & Wheeler, L. (1969). Model, need, and cost effects in helping behavior. *Journal of Personality and Social Psychology*, 12(2), 111–116. <https://doi.org/10.1037/h0027569>
- Wilhelm, M.O., & Bekkers, R. (2010). Helping Behavior, Dispositional Empathic Concern, and the Principle of Care. *Social Psychology Quarterly*, 73(1): 11–32. <https://doi.org/10.1177/0190272510361435>

Identifying replications

To identify replications, we first used two rule-based search strategies on Google Scholar. To keep the scope of our efforts within feasible bounds, we started our search for replications in studies indexed under the keyword “charitable behavior” because it produced the smallest number of results. Within this set, we followed an inclusive approach. We added the wildcard character * after the stem of replicate, searching for “replicat*”, so that we would find studies that contained “replicated”, “replicate”, “replicates”, “replicating”, and “replication”. Having learned from this search that a large proportion of search results merely mentioned replication but were not actual replications themselves, we used a more restrictive approach for the keyword “charitable giving”. We added only those replications that had not shown up in the previous search. Finally, we added replications we knew about that had not yet been identified through the rule-based searches. Figure A2 gives an overview of our search efforts.

Figure A2. Identified, screened, and included studies



Source: Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

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For all replication studies, we created the coding scheme below and assessed all studies. The resulting dataset is available at <https://osf.io/5thj9/files/zucw3>, and the code producing the results reported is at <https://osf.io/5thj9/files/t3puq>. The output is at <https://osf.io/5thj9/files/e9jqu>.

Replications of studies on “charitable behavior”

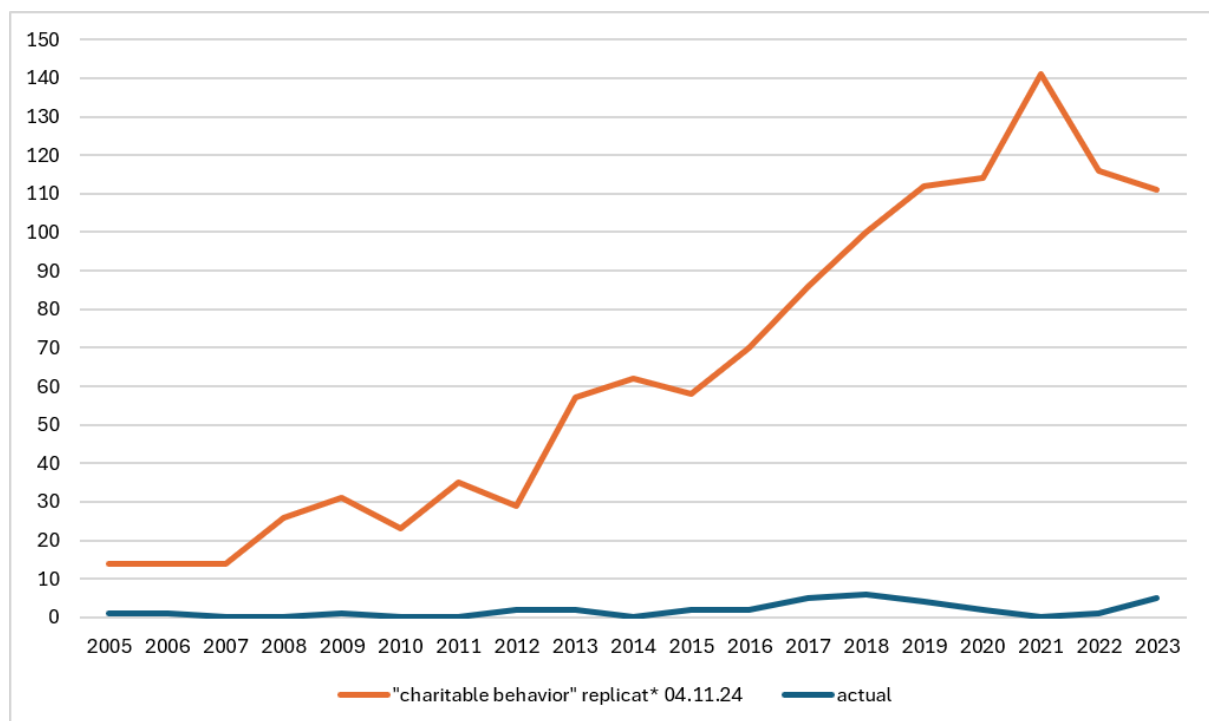
In November 2024, we (RB) searched Google Scholar for “replicat*” to identify terms related to replication in research reports on “charitable behavior” from the years 2005-2023, producing 1,213 results. Citation counts, bibliographic details of replications and further details are on tab “charitable behavior” replicat” of <https://osf.io/5thj9/files/jxkgs>.

We stratified the results by year of publication and screened them in four steps: 1) we manually inspected all titles and text fragments; 2) for studies that we deemed likely to repeat a previous study we visited the URLs and read the abstracts; 3) if the abstract suggested that the study repeated a previous study, we downloaded the full text; 4) we searched the full text and coded the meaning of the sentences containing “replicat*”.

Among the 1,213 search results we identified, 34 results actually replicated a previous study. We removed 2 working papers of which the published version also showed up in the search. In our main analysis we leave out 7 studies that did not analyze charitable giving behavior as a dependent variable, but instead examined intentions to give, or used it as an independent variable. Appendix B below lists the 25 unique results with prefix [A].

Mentions of terms related to replication increased strongly, but the number of actual replications hardly increased (see Figure A3). We interpret this pattern as a growing frequency of mentions as an increasing awareness of the importance of replication, and a growing gap with the actual practice of replication.

A3. Number of results in Google Scholar for “charitable behavior” + replicat and actual replications, 2000-2023*



The figure shows the number of search results in Google Scholar in November 2024 for “charitable behavior” and replication per year, and the number of actual replications. Source: <https://osf.io/5thj9/files/jxkgs>.

For each year, RB counted how terms related to replication were mentioned. For 1,179 search results on “charitable behavior” and “replicat*”, the abstracts and full text search results indicated that the studies did not repeat a previous study by other scholars. Across all years, about 31% of the search results were ‘internal replications’, building on previous findings by the same authors (“Study 2 partly replicates the findings of Study 1”). For about 15% of the search results terms related to replication were recommendations for future research, typically in the discussion section (“future research should replicate our findings”). For 6% of the search results terms related to replication appeared in the literature review section (“Previous replication attempts have been unsuccessful”). Five percent (5%) of search results were studies that did not measure charitable giving. Most of the other mentions occurred in sentences offering substantive examples (e.g., replication of DNA), discussion of the robustness of study findings (“subsample analyses replicating our main analyses”) or the application of study findings in practice (“our results may be replicated in fundraising campaigns”). For one third of the results (33%), the study title and text fragment did not contain content indicating that the study was repeating a previous study. For a subset of these results we looked up the full text to check the assumption that they were not replications, and found no studies that were replications. In the screening of search results we noticed that studies self-identifying as a replication in the title or abstract were much more likely to be replications than studies that mentioned terms related to replication elsewhere in the text.

Replications of studies on “charitable giving”

We (RB, HF, AG) repeated the steps above for “charitable giving”, in a refined search, adding only “replication” to get more relevant results. In November and December 2024, the search resulted in 17,300 search results. RB screened the first 15 pages containing 150 results, identified 20 replications, and coded them. In May 2026, the number of search results had increased to 18,500, and the rank order had changed. AG and HF screened the first 34 pages containing 340 results, identified 8 more replications of studies that qualified our criteria, and coded them. Appendix B lists them with prefix [B]. Due to time constraints, we had to stop screening the results after 34 pages. As a result, the review is not complete, and we may have missed some replications. We did notice that the number of actual replications declined strongly after page 30. Pages 30 - 34 yielded only 2 replications that satisfied our criteria and which we had not found previously.

Complementing the results from the two replicable search strategies, we also included 13 replications [C] that we knew about, but had not yet shown up in the search results.

Coding scheme

Variable	Label	Coding instructions
stID	study identifier	count indicator
URL	the URL of the DOI of the study	Look for URLs starting with https://doi.org/ . Avoid URLs unique to VU Amsterdam or your university library
DOI	the digital object identifier of the replication study	only the DOI number, without DOI:
pubyear	year of initial publication	year of online first / early publication if the study appeared online before it appeared in a printed edition
reference	bibliographic reference	copy and paste from Google Scholar, check page numbers
GScit	number of citations in Google Scholar	enter the number "cited by"
Studypub	type of publication	follow Item Type values in Zotero: 1 = Journal article, 2 = Thesis, 3 = Preprint or working paper, 4 = Conference Paper
whichstudy	for papers reporting multiple studies: which study was coded	enter the number of the study that you coded
empirical	the study presents new analyses of empirical data, either newly collected or existing data	Conceptual studies developing a theory or hypothesis and studies summarizing previous research without a new analysis are coded 0. For papers that do not report on empirical analyses the columns DV and following are not relevant, they should be left blank (not 0).
DV	the dependent variable is philanthropic behavior by individuals or households	Studies on corporate philanthropy and corporate social responsibility (CSR) are coded 0. Studies on giving intentions, hypothetical behavior, and studies on donations of substances of human origin (blood, organs) are coded 0. Studies on philanthropy as an independent variable are coded 0.
givobs	giving is observed by researchers or registered in a database	Studies in which giving behavior was a binding choice in experiments and surveys are coded 1, studies with register data in which donations are recorded in donor databases are coded 1, case studies in which researchers observed giving behavior are coded 1, studies with surveys in which participants self-reported giving behavior or intentions are coded 0

givselfrep	giving is self-reported by study participants	Studies with surveys in which participants self-reported giving behavior or intentions are coded 1
mech1	the study tests the effect of awareness of need	only if DV = 1; studies about the effects of information about recipients, the size of the problem addressed by the charity, number of victims. Studies on framing effects in this type of information are about psychological costs/benefits (mechanism 6)
mech2	the study tests the effect of solicitation	only if DV = 1; studies about the effects of various types of requests for donations, including the legitimization of paltry contributions (LPC), door-in-the-face (DITF), foot-in-the-mouth and other compliance strategies involving costs are about material costs (mechanism 3); studies about the social costs and benefits of giving are about reputation (mechanism 5); studies about the psychological costs and benefits of various compliance strategies are mechanism 6
mech3	the study tests the effect of material costs or benefits	only if DV = 1; studies about (tax) price effects, thank you gifts, rewards in crowdfunding campaigns
mech4	the study tests the effect of altruism	only if DV = 1; studies about altruism, i.e. how giving responds to the well-being of recipients in experiments, and changes in funding sources in experiments and annual reports (crowding in/crowding out). Studies testing relationships between giving and survey scales measuring altruistic values should be coded 0 if they do not measure actual changes in giving as a result of responses to the well-being of recipients. These studies test mechanism 7 (values)
mech5	the study tests the effect of reputation	only if DV = 1; studies about social costs and benefits of giving, i.e. watching eyes, public observability
mech6	the study tests the effect of psychological costs or benefits	only if DV = 1; studies about the (anticipated) psychological costs and benefits of giving: emotions such as empathy, shame, pride, guilt; cognitive biases such as gain/loss, rebate/matching framing effects; perceptions of ingroups and outgroups
mech7	the study tests the effect of values	only if DV = 1; studies about prosocial values, altruistic values, moral values, politics, world views
mech8	the study tests the effect of efficacy	only if DV = 1; studies about the effectiveness and impact of charities
experiment	the study has an experimental design	observations are allocated to treatment or control groups, in an online, lab, field or natural setting

cross	the study has a cross-sectional design	if experiment = 0: comparison of individuals or groups at one single point in time
panel	the study has a longitudinal design	if experiment = 0: comparison of the same individuals or groups at multiple points in time
survey	the study relies on data from a survey	note that many experiments also rely on surveys, e.g. to measure respondent characteristics, attention checks, and manipulation checks
studyn	number of observations in the study	including observations that were not used in the analysis due to missing data, excluded as outliers, or for other reasons
hypo	the study contains at least one hypothesis	the hypothesis is stated explicitly in the introduction or theory section of the paper or in the preregistration
hypo1text	H1 as formulated in the paper	copy the text of the hypothesis as it appears in the paper
H1stats	the test statistics of the first hypothesis test	copy the statistics of the first hypothesis test on charitable giving
H1n	the number of observations in the first hypothesis test	as reported in the text, or if not reported in the text, infer the number from a table of results or the number of degrees of freedom
replication	the study presents a replication	regardless of whether the replication concerns charitable giving as a DV; studies mentioning replication as worthwhile are coded 0
verification	the study reports on the same data with the same analysis as a previous study	only code if replication = 1
robustness	the study reports on the same data with a different analysis as a previous study	only code if replication = 1
reproduction	the study reports on different data with the same analysis as a previous study	only code if replication = 1
generalization	the study reports on different data with a different analysis as a previous study	only code if replication = 1

targetref	bibliographic reference of the original study which was replicated	Look for URLs starting with https://doi.org/ . Avoid URLs unique to VU Amsterdam or other university libraries
targetDOI	DOI of the original study which was replicated	only the DOI number, without DOI:
targetcit	number of citations of original study in Google Scholar	enter the number "cited by"
targetn	number of observations in the study that was replicated	as reported in the text, or if not reported in the text, infer the number from a table of results or the number of degrees of freedom
prereg	the replication mentions that it has a pre-registration	code any pre-registration as 1, including OSF, As Predicted, or ResearchBox
preregdev	the replication report deviates from the pre-registration	code any deviation as 1
datashared	the source data of the replication are available	the article itself or website provides a URL where the data are actually available
dataURL	the replication report contains a reference to a URL where the source data are available	e.g., "data for this study are publicly available at www.osf.io/ "
dataref	the bibliography of the replication contains a reference to the dataset	code only references to datasets as 1. code references to codebooks, user manuals and publications about the data as 0
datadesc	the reference to the source data is a publication describing the data	code references to codebooks, user manuals and publications about the data as 1
dataname	the reference to the source data is to the name of the dataset	e.g., "we used data from the Giving in the Netherlands Panel Survey"

codeshared	the replication report or repository shares the code producing the results reported	the article itself or website provides a URL where the code is actually available
coderuns	the shared code is executable	if codeshare = 1
publishedversion		only if unpublished: URL of published version
remark		any noteworthy things you see

Appendix C. Replication studies on charitable giving from three sources

[A] Google Scholar search for “replicat*” + “charitable behavior”

[B] Google Scholar search for “replication” + “charitable giving”

[C] other sources

[C] Adena, M., & Huck, S. (2019). Giving once, giving twice: A two-period field experiment on intertemporal crowding in charitable giving. *Journal of Public Economics*, 172, 127-134.

<https://doi.org/10.1016/j.jpubeco.2019.01.002>

[B] Ahumada, B., Chen, Y., Gupta, N., Hyde, K., Lepper, M., Mathews, W., Silveus, N., Vesterlund, L., Weidman, T., Wilson, A., Winichakul, K. P., & Zhou, L. (2022). Well Excuse Me! Replicating and Connecting Excuse-Seeking Behaviors. Smith College, Economics Faculty Publications.

https://scholarworks.smith.edu/eco_facpubs/103

[C] Almunia, M., Guceri, I., Lockwood, B., & Scharf, K. (2020). More giving or more givers? The effects of tax incentives on charitable donations in the UK. *Journal of Public Economics*, 183, 104114.

<https://doi.org/10.1016/j.jpubeco.2019.104114>

[A] Andreoni, J., Rao, J. M., & Trachtman, H. (2017). Avoiding the ask: A field experiment on altruism, empathy, and charitable giving. *Journal of Political Economy*, 125(3), 625-653.

<https://doi.org/10.1086/691703>

[B] Apinunmahakul, A., Barham, V., & Devlin, R. A. (2009). Charitable Giving, Volunteering, and the Paid Labor Market. *Nonprofit and Voluntary Sector Quarterly*, 38(1), 77-94.

<https://doi.org/10.1177/0899764008315845>

[B] Bakija, J. (2000). Distinguishing Transitory and Permanent Price Elasticities of Charitable Giving with Pre-Announced Changes in Tax Law. Department of Economics Working Papers, Department of Economics Working Papers, Article 2000-06. <https://ideas.repec.org/p/wil/wileco/2000-06.html>

[B] Bakija, J., & McClelland, R. (2005). Timing vs. Long-run Charitable Giving Behavior: Reconciling Divergent Approaches and Estimates (Working Paper Nos. 2005-08; Department of Economics Working Papers). Department of Economics, Williams College.

<https://EconPapers.repec.org/RePEc:wil:wileco:2005-08>

[C] Bekkers, R. (2011). Do Two Eyes See More Generosity Than One? Watching Eyes, Gender, and Giving. Paper presented at the 14th International Conference on Social Dilemmas, Amsterdam, July 7, 2011.

https://osf.io/preprints/psyarxiv/pseu2_v1

[B] Bekkers, R. (2015). When and Why Matches are More Effective Subsidies Than Rebates. Pp. 183-211 in Deck, C. Fatas, E., & Rosenblat, T. (Eds.). *Research in Experimental Economics, Volume 18: Replication in Economic Experiments*. Emerald Group Publishing. <https://doi.org/10.1108/S0193-230620150000018007>

- [C] Bruine de Bruin, W., & Ulqinaku, A. (2021). Effect of mortality salience on charitable donations: Evidence from a national sample. *Psychology and Aging*, 36(4), 415–420. <https://doi.org/10.1037/pag0000478>
- [C] Charles, C., Sloan, M. F., & Schubert, P. (2020). If someone else pays for overhead, do donors still care? *American Review of Public Administration*, 50(4-5), 415-427. <https://doi.org/10.1177/0275074020913989>
- [B] Costello, J. P., & Malkoc, S. A. (2022). Why Are Donors More Generous with Time Than Money? The Role of Perceived Control over Donations on Charitable Giving. *Journal of Consumer Research*, 49(4), 678–696. <https://doi.org/10.1093/jcr/ucac011>
- [B] Damgaard, M. T., & Gravert, C. (2018). The hidden costs of nudging: Experimental evidence from reminders in fundraising. *Journal of Public Economics*, 157, 15–26. <https://doi.org/10.1016/j.jpubeco.2017.11.005>
- [B] Dannenberg, A., Johansson-Stenman, O., & Wetzels, H. (2022). Status for the good guys: An experiment on charitable giving. *Economic Inquiry*, 60(2), 721–740. <https://doi.org/10.1111/ecin.13039>
- [A] Davis, D. D., Millner, E. L., & Reilly, R. J. (2005). Subsidy schemes and charitable contributions: a closer look. *Experimental Economics*, 8, 85-106. <https://doi.org/10.1007/s10683-005-0867-y>
- [A] Desai, R. M., & Kharas, H. The Wisdom of Crowd-funders: What motivates cross-border development aid? Global Economy and Development Working Paper 64. https://ciaotest.cc.columbia.edu/wps/bi/0030384/f_0030384_24557.pdf
- [B] Dickert, S., Kleber, J., Västfjäll, D., & Slovic, P. (2016). Mental Imagery, Impact, and Affect: A Mediation Model for Charitable Giving. *PLoS One*, 11(2), e0148274. <https://doi.org/10.1371/journal.pone.0148274>
- [A] Diederich, J., Eckel, C. C., Epperson, R., Goeschl, T., & Grossman, P. J. (2022). Subsidizing unit donations: Matches, rebates, and discounts compared. *Experimental Economics*, 25(2), 734-758. <https://doi.org/10.1007/s10683-021-09732-9>
- [A] Dur, R., & van Lent, M. (2018). Serving the public interest in several ways: Theory and empirics. *Labour Economics*, 51, 13-24. <https://doi.org/10.1016/j.labeco.2017.11.002>
- [A] Efendic, E., Bartoš, F., Vranka, M. A., & Bahník, Š. (2019). Unpacking the justifiability of dishonesty: Behavioral and process-tracing investigation. <https://psyarxiv.com/rn85h/download/?format=pdf>
- [A] Einolf, C. J. (2006). The roots of altruism: A gender and life course perspective. University of Virginia.
- [B] Ekström, M. (2012). Do watching eyes affect charitable giving? Evidence from a field experiment. *Experimental Economics*, 15(3), 530–546. <https://doi.org/10.1007/s10683-011-9312-6>
- [C] Fack, G., & Landais, C. (2010). Are tax incentives for charitable giving efficient? Evidence from France. *American Economic Journal: Economic Policy*, 2(2), 117-141. <https://doi.org/10.1257/pol.2.2.117>

- [B] Fang, Z., & Bian, Y. (2022). Another reason for the counterintuitive effects of thank-you gifts on charitable giving. *Frontiers in Psychology*, 13, 908556. <https://doi.org/10.3389/fpsyg.2022.908556>
- [A] Forbes, R. (2019). Alumni Giving: How Does Gender Matter? Lycoming College honors thesis. https://www.lycoming.edu/library/archives/honorspdfs/forbes_rebecca2019.pdf
- [B] Gandullia, L., Lezzi, E., & Parciasepe, P. (2020). Replication with MTurk of the experimental design by Gangadharan, Grossman, Jones & Leister (2018): Charitable giving across donor types. *Journal of Economic Psychology*, 78, 102268. <https://doi.org/10.1016/j.joep.2020.102268>
- [B] Greenway, T. S., Schnitker, S. A., & Shepherd, A. M. (2018). Can Prayer Increase Charitable Giving? Examining the Effects of Intercessory Prayer, Moral Intuitions, and Theological Orientation on Generous Behavior. *The International Journal for the Psychology of Religion*, 28(1), 3–18. <https://doi.org/10.1080/10508619.2017.1406790>
- [C] Grieder, M., Schmitz, J., & Schubert, R. (2021). Asking to give: moral licensing and pro-social behavior in the aggregate. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3920355
- [B] Huang, Q., Li, D., Zhou, C., Xu, Q., Li, P., & Warren, C. M. (2021). Multivariate pattern analysis of electroencephalography data reveals information predictive of charitable giving. *NeuroImage*, 242, 118475. <https://doi.org/10.1016/j.neuroimage.2021.118475>
- [B] Huck, S., & Rasul, I. (2011). Matched fundraising: Evidence from a natural field experiment. *Journal of Public Economics*, 95(5–6), 351–362. <https://doi.org/10.1016/j.jpubeco.2010.10.005>
- [A] Janssen, W. & Valkenburg, S. J. (2016). Trust and Charitable Behavior: A Study about the Effects of Trust on Charitable Behavior in the Netherlands. Bachelor's thesis, Utrecht University. <https://studenttheses.uu.nl/bitstream/handle/20.500.12932/23781/Valkenburg.%20S.%20-%20Scriptie.pdf?sequence=2>
- [A] Jung, M. H., Nelson, L. D., Gneezy, U., & Gneezy, A. (2017). Signaling virtue: Charitable behavior under consumer elective pricing. *Marketing Science*, 36(2), 187-194. <https://doi.org/10.1287/mksc.2016.1018>
- [A] Jung, M. H., Saccardo, S., Gneezy, A., & Nelson, L. D. (2023). Image concerns and generosity: field evidence and predictions. <http://dx.doi.org/10.2139/ssrn.4323906>
- [B] Kammermann, B., Türkay, S., Johnson, D., & Tobin, S. J. (2024). Do videogame rewards influence players' subsequent prosocial engagement? A preregistered partial replication study on the role of reward and reasoning. *International Journal of Human-Computer Studies*, 181, 103143. <https://doi.org/10.1016/j.ijhcs.2023.103143>
- [B] Kessler, J. B., & Meier, S. (2014). Learning from (failed) replications: Cognitive load manipulations and charitable giving. *Journal of Economic Behavior & Organization*, 102, 10–13. <https://doi.org/10.1016/j.jebo.2014.02.005>

- [B] Kim, M. (2013). Are Charitable Giving and Religious Attendance Complements or Substitutes? The Role of Measurement Error. *Journal of Policy Analysis and Management*, 32(2), 373–390.
<https://doi.org/10.1002/pam.21687>
- [B] Livingston, J. A., & Rasulmukhamedov, R. (2023). On the Interpretation of Giving in Dictator Games When the Recipient is a Charity. *Journal of Economic Behavior & Organization*, 208, 275–285.
<https://doi.org/10.1016/j.jebo.2023.01.025>
- [A] Lopez, J. J., Mohapatra, R. P., & Prew, P. (2017). Regional Patterns of Giving in the United States during the Great Recession. *International Journal of Culture and History*, 3 (4): 219, 226.
https://www.researchgate.net/profile/Paul-Prew/publication/322597818_Regional_Patterns_of_Giving_in_the_United_States_during_the_Great_Recession/links/61dc63c7323a2268f99631e8/Regional-Patterns-of-Giving-in-the-United-States-during-the-Great-Recession.pdf
- [A] Luccasen, A. & Grossman, P.J. (2013). A robust test of warm glow giving and spiteful pleasure in a “real donation” experiment with and without earned endowments. Monash University Department of Economics Discussion Paper 29/13.
<https://www.monash.edu/business/economics/research/publications/2013/2913robustlucasengrossman.pdf>
- [A] Marcinkiewicz, M. (2016). Increasing charitable donations: The limits of the unit asking effect. Villanova University. <https://www.proquest.com/openview/858f5aa23a41a1ec479bfb9388012098/>
- [A] Margolis, M.F., & Sances, M.W. (2017). Partisan differences in nonpartisan activity: The case of charitable giving. *Political Behavior*, 39, 839-864. <https://doi.org/10.1007/s11109-016-9382-4>
- [A] Mayorga, M.W. (2019). An Introspection Intervention for Perceived Inefficacy in Charitable Giving. Doctoral dissertation, University of Oregon.
https://scholarsbank.uoregon.edu/xmlui/bitstream/handle/1794/24903/Mayorga_oregon_0171A_12496.pdf?sequence=1&isAllowed=y
- [B] Murphy, J. J., Batmunkh, N., Nilsson, B., & Ray, S. (2015). The Impact of Social Information on the Voluntary Provision of Public Goods: A Replication Study. Pp. 41–50 in: Deck, C. Fatas, E., & Rosenblat, T. (Eds.). *Research in Experimental Economics, Volume 18: Replication in Economic Experiments*. Emerald Group Publishing. <https://doi.org/10.1108/S0193-230620150000018002>
- [A] Naeem, S., & Zaman, A. (2016). Charity and gift exchange: Cultural effects. *Voluntas*, 27, 900-919.
<https://doi.org/10.1007/s11266-015-9655-2>
- [C] Newman, G. E., & Shen, Y. J. (2012). The counterintuitive effects of thank-you gifts on charitable giving. *Journal of Economic Psychology*, 33(5), 973-983. <https://doi.org/10.1016/j.joep.2012.05.002>
- [B] Oda, R., & Ichihashi, R. (2016). Effects of Eye Images and Norm Cues on Charitable Donation: A Field Experiment in an Izakaya. *Evolutionary Psychology*, 14(4), 1474704916668874.
<https://doi.org/10.1177/1474704916668874>

- [A] Rahwan, Z., Hauser, O. P., Kochanowska, E., & Fasolo, B. (2018). High stakes: A little more cheating, a lot less charity. *Journal of Economic Behavior & Organization*, 152, 276-295. <https://doi.org/10.1016/j.jebo.2018.04.021>
- [C] Roberts, J. R., & Maxfield, M. (2019). Mortality salience and age effects on charitable donations. *American Behavioral Scientist*, 63(14), 1863-1884. <https://doi.org/10.1177/0002764219850864>
- [C] Sasaki, S., Kurokawa, H., & Ohtake, F. (2022). An experimental comparison of rebate and matching in charitable giving: The case of Japan. *Japanese Economic Review*, 73(1), 147-177. <https://doi.org/10.1007/s42973-021-00085-9>
- [A] Saunders, T. J., Taylor, A. H., & Atkinson, Q. D. (2016). No evidence that a range of artificial monitoring cues influence online donations to charity in an MTurk sample. *Royal Society Open Science*, 3(10), 150710. <https://royalsocietypublishing.org/doi/pdf/10.1098/rsos.150710>
- [B] Schervish, P. G., Coutsoukis, P. E., & Havens, J. J. (1998). Social participation and charitable giving revisited: Replication of a multivariate analysis. Annual Meeting of the Association for Research on Nonprofit Organizations and Voluntary Action, 2-4. <https://www.academia.edu/download/79874739/socialparticipation.pdf>
- [B] Schmukle, S. C., Korndörfer, M., & Egloff, B. (2019). No evidence that economic inequality moderates the effect of income on generosity. *Proceedings of the National Academy of Sciences*, 116(20), 9790-9795. <https://doi.org/10.1073/pnas.1807942116>
- [B] Schönegger, P., & Wagner, J. (2019). The moral behavior of ethics professors: A replication-extension in German-speaking countries. *Philosophical Psychology*, 32(4), 532-559. <https://doi.org/10.1080/09515089.2019.1587912>
- [A] Sekścińska, K., Trzcińska, A., & Maison, D. (2023). Charity campaigns with promotion-framed goals are more effective than those with prevention-framed goals. *PLoS One*, 18(8), e0286028. <https://doi.org/10.1371/journal.pone.0286028>
- [B] Simpson, B., Irwin, K., & Lawrence, P. (2006). Does a "Norm of Self-Interest" Discourage Prosocial Behavior? Rationality and Quid Pro Quo in Charitable Giving. *Social Psychology Quarterly*, 69(3), 296-306. <https://doi.org/10.1177/019027250606900306>
- [B] Sisco, M. R., & Weber, E. U. (2019). Examining charitable giving in real-world online donations. *Nature Communications*, 10(1), 3968. <https://doi.org/10.1038/s41467-019-11852-z>
- [B] Smith, R. W., Faro, D., & Burson, K. A. (2013). More for the Many: The Influence of Entitativity on Charitable Giving. *Journal of Consumer Research*, 39(5), 961-976. <https://doi.org/10.1086/666470>
- [A] Sparks, A. M., Fessler, D. M., & Holbrook, C. (2019). Elevation, an emotion for prosocial contagion, is experienced more strongly by those with greater expectations of the cooperativeness of others. *PLoS One*, 14(12), e0226071. <https://doi.org/10.1371/journal.pone.0226071>

- [B] Spiteri, G. W. (2022). Does the evaluability bias hold when giving to animal charities? *Judgment and Decision Making*, 17(2), 315–330. <https://doi.org/10.1017/S1930297500009128>
- [A] Stinn, J. L. (2018). Donor Attitudes Toward Overhead Costs and the Effects on Nonprofit Organizations. Doctoral dissertation, The Florida State University.
<https://diginole.lib.fsu.edu/islandora/object/fsu:647306/datastream/PDF/view>
- [A] Thomas-Walters, L., & J Raihani, N. (2017). Supporting conservation: The roles of flagship species and identifiable victims. *Conservation Letters*, 10(5), 581-587. <https://doi.org/10.1111/conl.12319>
- [C] Van Teunenbroek, P.S.C. & Bekkers, R. (2019). Follow the Crowd: Social Information and Crowdfunding Donations in a Large Field Experiment. *Journal of Behavioral Public Administration*, 3(1): 1-17. <https://doi.org/10.30636/jbpa.31.87>
- [A] Verhaert, G. A., & Van den Poel, D. (2012). The role of seed money and threshold size in optimizing fundraising campaigns: Past behavior matters!. *Expert Systems with Applications*, 39(18), 13075-13084. <https://doi.org/10.1016/j.eswa.2012.04.088>
- [C] Vu, L., Molho, C., Soraperra, I., Fiedler, S., & Shalvi, S. (2024). Giving (in) to help an identified person. *Journal of Experimental Social Psychology*, 110, 104557. <https://doi.org/10.1016/j.jesp.2023.104557>
- [C] Yörük, B. K. (2010). Charitable giving by married couples revisited. *Journal of Human Resources*, 45(2), 497-516. <https://doi.org/10.3368/jhr.45.2.497>
- [A] Zhang, L. (2023). Combining Experiments and Structural Econometrics to Understand Motives for Charitable Giving. The University of Alabama.
<https://ir.ua.edu/bitstreams/e6f46f87-23fe-4fa2-89d3-f9c2a8ccc838/download>

Appendix D. Further Exploratory Analyses

Small study bias. Studies with a small number of observations can produce strong effects, purely as a result of chance, while in reality no reliable effect exists. Small samples favor positive results (Loken & Gelman, 2017). Many studies in the social and behavioral sciences are underpowered (Ioannidis, Stanley & Doucouliagos, 2017; Arel-Bundock et al., 2022; Maxwell, 2004). Replications tend to have larger numbers of observations (Tyner et al., 2026).

The median number of observations in 59 replicated studies for which we could find this information was 405. We find no evidence that a smaller number of observations is associated with replications finding different results than preceding studies. In fact, the median number of observations in 36 replications that produced results in the same direction as the replicated study was 248, lower than the median number of observations of 573 in 23 replications that produced null findings or results running counter to the replicated study. This result is due to differences between experiments and other studies. Results of the 32 replications of results from 48 experiments were in line with the replicated study (63%). Of the 18 replications with a non-experimental design, 9 results were in line with previous findings (50%). The difference is not significant ($\chi^2 = 3.825$, $df = 1$, $p = .050$).

The median number of observations in 42 replicated experiments was 181, in the 18 other studies the median number of observations was 3572. Median regression indicates that the number of observations in replicated studies are not associated with results of replications having similar or different results, when the difference in study designs is taken into account (coefficient: -117 (SE: 1410.279), $t = -0.08$, $p = .934$).

Citation counts. A commonly used proxy for research quality is the number of citations that a study has received, which partly reflects the impact factor of the journal that published the study. However, more highly cited research is not more replicable than less cited research (Youyou, Yang, & Uzzi, 2023). In fact, studies in the Open Science Collaboration (2015) that were not successfully replicated were cited more often than studies that did replicate well (Serra-Garcia & Gneezy, 2021).

We find no evidence that citation counts differ between replicated studies that replications found similar results for and replications with null-findings or negative results. The median number of citations in Google Scholar for 39 previous studies that replications found similar results for was 27; for the 27 other studies the median number of citations to the previous study was 33. The median number of citations to experiments (27) was similar to that of studies with non-experimental designs (33). Median regression indicates that the number of citations for replicated studies is not different for replications having similar or different results, when the difference in study designs is taken into account (coefficient: 8 (SE: 16.570), $t = 0.48$, $p = .630$).

Changes after peer review. A common questionable research practice is ‘harking’, hypothesizing after results are known (Kerr, 1998). A recent study in Germany suggests that about half of published hypotheses in the social, behavioral and economic studies analyzing large population survey data are formulated *after* results are known (Poppa et al., 2025). A study in the field of management research (O’Boyle, Banks & Gonzalez-Mulé, 2017) suggests that peer review - especially at journals ranked as ‘more prestigious’ - introduces questionable research practices such as adding or deleting data points, covariates, and hypotheses.

In our set of replications, most studies that were available as a working paper or preprint did not change substantially when they were published. Sometimes dissertation advisors appear as co-authors on journal article publications (e.g., Portillo & Stinn, 2018). We found one piece of evidence for ‘hypothesizing after results are known’ as a result of peer review: Margolis & Sances (2017) support a hypothesis that was absent in the working paper (Margolis & Sances, 2015), while the empirical analyses did not change. The working paper stated that “partisanship and ideology may not determine overall levels of giving, [but] political differences may affect *where* partisans choose to donate” (emphasis in original). The published version states “we hypothesize that conservatives and Republicans will give more to charity than liberals and Democrats”.

References

- Arel-Bundock, V., Briggs, R. C., Doucouliagos, H., Mendoza Avaña, M., & Stanley, T. D. (2026). Quantitative Political Science Research is Greatly Underpowered. *Journal of Politics*, 88(1): 36-46. <https://doi.org/10.1086/734279>
- Kerr, N. L. (1998). HARKing: Hypothesizing after the results are known. *Personality and Social Psychology Review*, 2(3), 196-217. https://doi.org/10.1207/s15327957pspr0203_4
- Loken, E., & Gelman, A. (2017). Measurement error and the replication crisis. *Science*, 355(6325), 584-585. <https://doi.org/10.1126/science.aal3618>
- Margolis, M.F., & Sances, M.W. (2015). Partisan differences in charitable giving: Evidence from individual-level survey data. <https://www.michelemargolis.com/uploads/2/0/2/0/20207607/giving.pdf>
- Maxwell, S. E. (2004). The Persistence of Underpowered Studies in Psychological Research: Causes, Consequences, and Remedies. *Psychological Methods*, 9(2), 147–163. <https://doi.org/10.1037/1082-989X.9.2.147>
- O’Boyle Jr, E. H., Banks, G. C., & Gonzalez-Mulé, E. (2017). The chrysalis effect: How ugly initial results metamorphosize into beautiful articles. *Journal of Management*, 43(2), 376-399. <https://doi.org/10.1177/0149206314527133>
- Poppa, C., Nießen, D., Daikeler, J., Silber, H., Weiß, B., & Richter, D. (2025). The Tip of the Iceberg? Insights into the Prevalence of Publication Bias in Two Probability-Based Academic Panels. https://doi.org/10.31234/osf.io/bj3g9_v1
- Serra-Garcia, M., & Gneezy, U. (2021). Nonreplicable publications are cited more than replicable ones. *Science Advances*, 7(21), eabd1705. <https://doi.org/10.1126/sciadv.abd1705>